Entanglement Entropy in Conformal Field Theory

Replica Trick, Twist Fields, and the Formula $S = \frac{c}{3} \ln L$

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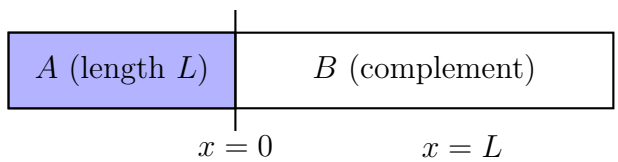
1 Introduction

Entanglement entropy is a fundamental measure of quantum correlations in many-body systems. In conformal field theory (CFT), it takes a universal logarithmic form that depends only on the central charge c, making it a powerful tool for characterizing quantum phases and critical points.

1.1 Physical Setup

Consider a 1+1D quantum system in its ground state $|0\rangle$. We partition the system into two regions:

- Region A: interval of length L (the subsystem we study)
- Region B: the complement (traced out)



1.2 Entanglement Entropy Definition

The reduced density matrix is obtained by tracing out region B:

$$\rho_A = \mathrm{Tr}_B \, |0\rangle \langle 0| \tag{1}$$

The von Neumann entanglement entropy is:

$$S_A = -\,\mathrm{Tr}(\rho_A \ln \rho_A) \tag{2}$$

1.3 Physical Interpretation

Value	**Interpretation**	**Example**
$S_A = 0$	Product state (no entanglement)	$\|\psi\rangle_A \otimes \|\phi\rangle_B$
$S_A > 0$	Entangled state	Bell state, GHZ state
$S_A \propto L$	Volume law (thermal states)	High-energy eigenstates
$S_A \propto \ln L$	Area law in 1D (CFT ground states)	Critical systems

1.4 Main Result

For a 1+1D CFT with central charge c, the entanglement entropy of an interval of length L is:

$$S_A = \frac{c}{3} \ln \left(\frac{L}{a} \right) + \text{const} \tag{3}$$

where a is the UV cutoff (lattice spacing).

For the Ising model ($c = 1/2$):

$$S_A = \frac{1}{6} \ln \left(\frac{L}{a} \right) + \text{const} \tag{4}$$

2 The Replica Trick: Avoiding Direct Diagonalization

2.1 The Fundamental Challenge

The entanglement entropy is defined as:

$$S_A = -\operatorname{Tr}(\rho_A \ln \rho_A) \tag{5}$$

To compute this directly, we face a formidable obstacle:

- **Step 1:** Diagonalize the reduced density matrix ρ_A to find all eigenvalues $\{\lambda_i\}$
- **Step 2:** Compute the sum: $S_A = -\sum_i \lambda_i \ln \lambda_i$

Why is this hard?
For a quantum field theory or many-body system:

- The Hilbert space dimension grows *exponentially* with subsystem size: $\dim(\mathcal{H}_A) \sim d^L$
- ρ_A is typically a dense, high-dimensional matrix
- Finding eigenvalues of an exponential-sized matrix is computationally intractable
- Even if we could diagonalize ρ_A, the eigenvalues themselves are complicated functionals of the field configuration

Example: For a chain of $L = 100$ spin-1/2 particles:

$$\dim(\mathcal{H}_A) = 2^{100} \approx 10^{30} \tag{6}$$

Diagonalizing a $10^{30} \times 10^{30}$ matrix is impossible!

2.2 Rényi Entropies

Instead of attacking S_A directly, we compute a family of related quantities.

Definition: The **Rényi entropies** are:

$$S_A^{(n)} = \frac{1}{1-n} \ln \operatorname{Tr}(\rho_A^n) \tag{7}$$

where n is a positive integer ($n = 2, 3, 4, \ldots$).

In path integral language, $\operatorname{Tr}(\rho_A^n)$ has a geometric interpretation:

- $\operatorname{Tr}(\rho_A^n)$ involves *multiplying* density matrices, not diagonalizing them
- We can compute $\operatorname{Tr}(\rho_A^n)$ for integer n, then analytically continue to real n
- The limit $n \to 1$ recovers the von Neumann entropy

$\mathrm{Tr}(\rho_A^n)$ is called the *purity* (for $n = 2$) or *Rényi purity* (for general n).

Proof: Let $\{\lambda_i\}$ be the eigenvalues of ρ_A. Since ρ_A is a density matrix:

$$\sum_i \lambda_i = 1, \quad 0 \leq \lambda_i \leq 1 \tag{8}$$

For $n = 2$:

$$\mathrm{Tr}\left(\rho_A^2\right) = \sum_i \lambda_i^2 \tag{9}$$

Pure state: A pure state satisfies $\rho_A = |\psi\rangle\langle\psi|$, which is a rank-1 projector:

$$\rho_A^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \rho_A \tag{10}$$

Taking the trace:

$$\mathrm{Tr}\left(\rho_A^2\right) = \mathrm{Tr}(\rho_A) = 1 \tag{11}$$

In the eigenbasis, this corresponds to one eigenvalue $\lambda_1 = 1$ and all others $\lambda_{i>1} = 0$.

Mixed state: A mixed state is a convex combination of pure states:

$$\rho_A = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad \sum_i p_i = 1, \quad 0 < p_i < 1 \tag{12}$$

The eigenvalues are $\lambda_i = p_i$. Since $0 < \lambda_i < 1$ for at least two eigenvalues:

$$\lambda_i^2 < \lambda_i \quad \text{(strict inequality for each } 0 < \lambda_i < 1) \tag{13}$$

Summing over all eigenvalues:

$$\mathrm{Tr}\left(\rho_A^2\right) = \sum_i \lambda_i^2 < \sum_i \lambda_i = 1 \tag{14}$$

The inequality is strict because at least two eigenvalues contribute.

Entanglement measure: For a system entangled with environment, ρ_A is mixed. More entanglement $\Rightarrow$ more uniform eigenvalue distribution $\Rightarrow$ smaller $\mathrm{Tr}(\rho_A^n)$:

- $\mathrm{Tr}(\rho_A^2) = 1$: pure state (no entanglement)
- $\mathrm{Tr}(\rho_A^2) < 1$: mixed state (entangled with environment)
- Smaller $\mathrm{Tr}(\rho_A^n) \Rightarrow$ more entanglement

2.3 Recovering von Neumann Entropy

The key claim is:

$$S_A = \lim_{n \to 1} S_A^{(n)} \tag{15}$$

Let $\{\lambda_i\}$ be the eigenvalues of ρ_A. Then:

$$\mathrm{Tr}(\rho_A^n) = \sum_i \lambda_i^n \tag{16}$$

Define $f(n) = \ln \mathrm{Tr}(\rho_A^n) = \ln\left(\sum_i \lambda_i^n\right)$.
The Rényi entropy is:

$$S_A^{(n)} = \frac{f(n)}{1-n} \tag{17}$$

Taking $n \to 1$, we get $0/0$ indeterminate form. Apply L'Hôpital's rule:

$$S_A = \lim_{n\to 1} \frac{f(n)}{1-n} = -\left.\frac{df}{dn}\right|_{n=1} \tag{18}$$

Compute the derivative:

$$\frac{df}{dn} = \frac{d}{dn} \ln\left(\sum_i \lambda_i^n\right) \tag{19}$$

$$= \frac{\sum_i \lambda_i^n \ln \lambda_i}{\sum_i \lambda_i^n} \tag{20}$$

At $n = 1$:

$$\left.\frac{df}{dn}\right|_{n=1} = \frac{\sum_i \lambda_i \ln \lambda_i}{\sum_i \lambda_i} = \sum_i \lambda_i \ln \lambda_i = -S_A \tag{21}$$

Therefore:

$$S_A = -(-S_A) = S_A \tag{22}$$

2.4 Physical Picture

To compute $\mathrm{Tr}(\rho_A^n)$:

1. Take n *copies* (replicas) of the system
2. Sew them together in a cyclic pattern
3. Compute a partition function on this n-sheeted geometry
4. Take $n \to 1$ at the end

The name "replica" comes from step 1.
Historical note: This method was borrowed from spin glass theory.

2.5 Workflow

1. **Compute** $\mathrm{Tr}(\rho_A^n)$ for integer $n \geq 2$
   - Use path integral representation
   - Geometric interpretation: n-sheeted Riemann surface
2. **Analytically continue** to real n
   - Treat n as a continuous parameter
   - Assume smooth behavior near $n = 1$
3. **Take the limit** $n \to 1$
   - Apply L'Hôpital's rule
   - Recover von Neumann entropy

3 Path Integral Representation

3.1 Trace Formula in Field Basis

The trace of ρ_A^n can be expressed as a functional integral over field configurations in region A.

Statement:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A \, \langle \phi_A | \rho_A^n | \phi_A \rangle \tag{23}$$

where:

- Tr = Trace over Hilbert space $\mathcal{H}_A$
- $\int \mathcal{D}\phi_A$ = Functional integral over all field configurations in region A
- $|\phi_A\rangle$ = Field eigenstate in region A

Proof:

Step 1: Definition of trace. In any orthonormal basis $\{|k\rangle\}$ of $\mathcal{H}_A$:

$$\mathrm{Tr}(\rho_A^n) = \sum_k \langle k | \rho_A^n | k \rangle \tag{24}$$

Step 2: Choose field eigenstate basis. Field eigenstates in region A:

$$\hat{\phi}(x)|\phi_A\rangle = \phi_A(x)|\phi_A\rangle \quad \text{for } x \in A \tag{25}$$

Step 3: Completeness relation. For field eigenstates:

$$\int \mathcal{D}\phi_A \, |\phi_A\rangle\langle\phi_A| = I_A \quad \text{(identity on } \mathcal{H}_A\text{)} \tag{26}$$

The measure $\mathcal{D}\phi_A$ is defined by discretizing space:

$$A = \{x_1, x_2, \ldots, x_M\}, \quad \phi_A = \{\phi_1, \phi_2, \ldots, \phi_M\} \tag{27}$$

$$\mathcal{D}\phi_A = \prod_{j=1}^{M} d\phi_j \quad \xrightarrow{M\to\infty} \quad \text{functional measure} \tag{28}$$

Step 4: Orthogonality. Field eigenstates are orthogonal:

$$\langle \phi_A | \phi_A' \rangle = \delta[\phi_A - \phi_A'] = \prod_{x\in A} \delta(\phi_A(x) - \phi_A'(x)) \tag{29}$$

Step 5: Insert completeness into trace:

$$\mathrm{Tr}(\rho_A^n) = \sum_k \langle k | \rho_A^n \left(\int \mathcal{D}\phi_A \, |\phi_A\rangle\langle\phi_A| \right) |k\rangle \tag{30}$$

$$= \int \mathcal{D}\phi_A \sum_k \langle k | \rho_A^n | \phi_A \rangle \langle \phi_A | k \rangle \tag{31}$$

Step 6: Swap sum and integral. Using $\sum_k |k\rangle\langle k| = I_A$:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A \sum_k \langle\phi_A|k\rangle\langle k|\rho_A^n|\phi_A\rangle \tag{32}$$

$$= \int \mathcal{D}\phi_A \, \langle\phi_A| \left(\sum_k |k\rangle\langle k|\right) \rho_A^n|\phi_A\rangle \tag{33}$$

$$= \int \mathcal{D}\phi_A \, \langle\phi_A|\rho_A^n|\phi_A\rangle \tag{34}$$

Step 7: Final result:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A \, \langle\phi_A|\rho_A^n|\phi_A\rangle \tag{35}$$

3.2 Alternative Derivation: Direct from Matrix Elements

In the field representation, ρ_A^n has matrix elements:

$$(\rho_A^n)(\phi_A, \phi'_A) = \langle\phi_A|\rho_A^n|\phi'_A\rangle \tag{36}$$

For a discrete basis, the trace is the sum of diagonal elements:

$$\mathrm{Tr}\left(\hat{O}\right) = \sum_i O_{ii} \tag{37}$$

For a continuous basis (field configurations), the sum becomes a functional integral:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A \, \langle\phi_A|\rho_A^n|\phi_A\rangle \tag{38}$$

3.3 Lattice Regularization

To make the functional integral well-defined, we discretize space.

Lattice setup:

- Region A: M lattice sites $\{x_1, x_2, \ldots, x_M\}$
- Field configuration: $\vec{\phi} = (\phi_1, \phi_2, \ldots, \phi_M)$ where $\phi_j = \phi(x_j)$
- Hilbert space: $\mathcal{H}_A = L^2(\mathbb{R}^M)$ (square-integrable functions of $\vec{\phi}$)

Field eigenstates on lattice:

$$|\vec{\phi}\rangle \quad \text{where} \quad \hat{\phi}_j|\vec{\phi}\rangle = \phi_j|\vec{\phi}\rangle \tag{39}$$

Completeness:

$$\int d^M\phi \, |\vec{\phi}\rangle\langle\vec{\phi}| = I_A, \quad d^M\phi = d\phi_1 \, d\phi_2 \cdots d\phi_M \tag{40}$$

Orthogonality:

$$\langle\vec{\phi}|\vec{\phi}'\rangle = \delta^M(\vec{\phi}-\vec{\phi}') = \prod_{j=1}^{M}\delta(\phi_j-\phi_j') \tag{41}$$

Trace on lattice:

$$\begin{aligned}
\mathrm{Tr}(\rho_A^n) &= \sum_k \langle k|\rho_A^n|k\rangle && (42)\\
&= \sum_k \langle k|\rho_A^n \left(\int d^M\phi\, |\vec{\phi}\rangle\langle\vec{\phi}|\right)|k\rangle && (43)\\
&= \int d^M\phi \sum_k \langle\vec{\phi}|k\rangle\langle k|\rho_A^n|\vec{\phi}\rangle && (44)\\
&= \int d^M\phi\, \langle\vec{\phi}|\rho_A^n|\vec{\phi}\rangle && (45)
\end{aligned}$$

Continuum limit ($M\to\infty$):

$$\int d^M\phi \to \int \mathcal{D}\phi_A, \quad |\vec{\phi}\rangle \to |\phi_A\rangle \tag{46}$$

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A\, \langle\phi_A|\rho_A^n|\phi_A\rangle \tag{47}$$

3.4 Euclidean Path Integral for Ground State

Ground state preparation:

$$|0\rangle = \lim_{T\to\infty} e^{-HT}|\text{arbitrary}\rangle \tag{48}$$

In the path integral formalism, this corresponds to evolving from $\tau=-\infty$ to $\tau=0$.

Derivation:

Consider the transition amplitude from an arbitrary state $|\psi\rangle$ at $\tau=-T$ to a field configuration $|\phi_f\rangle$ at $\tau=0$:

$$\langle\phi_f|e^{-HT}|\psi\rangle = \int_{\phi(-T)=\psi}^{\phi(0)=\phi_f} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{49}$$

Expand $|\psi\rangle$ in the energy eigenbasis:

$$|\psi\rangle = c_0|0\rangle + \sum_{n>0} c_n|n\rangle \tag{50}$$

Apply Euclidean time evolution:

$$\begin{aligned}
e^{-HT}|\psi\rangle &= c_0 e^{-E_0T}|0\rangle + \sum_{n>0} c_n e^{-E_nT}|n\rangle && (51)\\
&= c_0 e^{-E_0T}\left[|0\rangle + \sum_{n>0}\frac{c_n}{c_0}e^{-(E_n-E_0)T}|n\rangle\right] && (52)
\end{aligned}$$

Since $E_n > E_0$ for all excited states, as $T \to \infty$:

$$e^{-(E_n - E_0)T} \to 0 \quad \text{for all } n > 0 \tag{53}$$

Therefore:

$$\lim_{T\to\infty} e^{-HT}|\psi\rangle = c_0 e^{-E_0 T}|0\rangle \tag{54}$$

The ground state is projected out. Normalizing:

$$|0\rangle = \lim_{T\to\infty} \frac{e^{-HT}|\psi\rangle}{\sqrt{\langle\psi|e^{-2HT}|\psi\rangle}} \tag{55}$$

Path integral representation:

$$\langle\phi_f|0\rangle = \frac{1}{\sqrt{Z}} \int_{\phi(-\infty)=\text{arbitrary}}^{\phi(0)=\phi_f} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{56}$$

The ground state wavefunctional is obtained by integrating over all field configurations from $\tau = -\infty$ to $\tau = 0$ with boundary condition $\phi(0) = \phi_f$.

3.4.1 Definition of the Functional Measure

What is $\mathcal{D}\phi$?

The functional measure $\mathcal{D}\phi$ (also called the path integral measure) is the continuum limit of a product of ordinary integrals over field values at each spacetime point.

Lattice regularization:

Discretize spacetime into a lattice with:

- Spatial sites: $x_j = ja$, $j = 1, \ldots, N_s$ (lattice spacing a)
- Time slices: $\tau_k = k\epsilon$, $k = 1, \ldots, N_\tau$ (time step ϵ)

The field becomes a set of discrete variables:

$$\phi(x, \tau) \quad \to \quad \phi_{j,k} \equiv \phi(x_j, \tau_k) \tag{57}$$

Functional measure as a product:

$$\mathcal{D}\phi \quad \equiv \quad \lim_{a\to 0,\, \epsilon\to 0} \prod_{j=1}^{N_s} \prod_{k=1}^{N_\tau} d\phi_{j,k} \tag{58}$$

Each $d\phi_{j,k}$ is an ordinary Lebesgue integral over the real line (assuming a real scalar field).

Normalization:

The measure includes a normalization factor to ensure convergence:

$$\mathcal{D}\phi = \mathcal{N} \prod_{j,k} d\phi_{j,k} \tag{59}$$

where $\mathcal{N}$ is chosen such that:

$$\int \mathcal{D}\phi\, e^{-S_E[\phi]} = Z \tag{60}$$

For a free scalar field, $\mathcal{N}$ is determined by Gaussian integration.

Why is this well-defined?

The functional integral is defined as the limit of well-defined finite-dimensional integrals:

$$\int \mathcal{D}\phi\, F[\phi] = \lim_{N_s, N_\tau \to \infty} \int \left(\prod_{j,k} d\phi_{j,k} \right) F(\{\phi_{j,k}\}) \tag{61}$$

For free (Gaussian) theories, the limit exists and can be computed exactly. For interacting theories, the measure is defined perturbatively or via lattice regularization.

Analogy with quantum mechanics:

For a single particle:

$$\int \mathcal{D}q(\tau) = \lim_{N \to \infty} \int dq_1 \cdots dq_{N-1} \tag{62}$$

where $q_k = q(\tau_k)$ and the endpoints q_0, q_N are fixed by boundary conditions.

In the replica trick:

The measure $\mathcal{D}\phi_A$ for the reduced density matrix trace is:

$$\int \mathcal{D}\phi_A = \int \prod_{x \in A} d\phi_A(x) \tag{63}$$

integrating over all field configurations in region A only.

3.5 Path Integral Derivation: Quantum Mechanics

System: Single particle in 1D with Hamiltonian:

$$H = \frac{\hat{p}^2}{2m} + V(\hat{q}) \tag{64}$$

Goal: Compute transition amplitude in Euclidean time:

$$\langle q_f | e^{-\beta H} | q_i \rangle \tag{65}$$

where $\beta = 1/T$ (inverse temperature).

3.5.1 Trotter Decomposition

Divide time into N slices:

$$\epsilon = \beta/N, \quad e^{-\beta H} = (e^{-\epsilon H})^N \tag{66}$$

Insert $N-1$ complete sets of position states:

$$\langle q_f | e^{-\beta H} | q_i \rangle = \int dq_1 \cdots dq_{N-1} \prod_{k=0}^{N-1} \langle q_{k+1} | e^{-\epsilon H} | q_k \rangle \tag{67}$$

where $q_0 = q_i$, $q_N = q_f$.

3.5.2 Short-Time Propagator

For small ϵ, use Trotter formula:

$$e^{-\epsilon H} = e^{-\epsilon(\hat{p}^2/(2m)+V(\hat{q}))} \approx e^{-\epsilon\hat{p}^2/(2m)} e^{-\epsilon V(\hat{q})} + O(\epsilon^2) \tag{68}$$

Insert momentum eigenstates:

$$\langle q_{k+1}|e^{-\epsilon H}|q_k\rangle \approx \int \frac{dp_k}{2\pi} \langle q_{k+1}|e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle\langle p_k|e^{-\epsilon V(\hat{q})}|q_k\rangle \tag{69}$$

Origin of the $1/(2\pi)$ factor:

The completeness relation for momentum eigenstates depends on the normalization convention. With $\langle p|q\rangle = e^{-ipq}$ (no prefactor):

$$\langle p|p'\rangle = 2\pi\delta(p-p') \tag{70}$$

The completeness relation is:

$$\int \frac{dp}{2\pi} |p\rangle\langle p| = I \tag{71}$$

The $1/(2\pi)$ ensures proper normalization. Equivalently, one can use $\langle p|q\rangle = \frac{1}{\sqrt{2\pi}} e^{-ipq}$ with $\int dp\, |p\rangle\langle p| = I$.

Derivation of matrix elements:

First factor: $\langle p_k|e^{-\epsilon V(\hat{q})}|q_k\rangle$

Lemma: If $V(\hat{q})$ is a function of the position operator, it acts diagonally on position eigenstates:

$$V(\hat{q})|q\rangle = V(q)|q\rangle \tag{72}$$

Proof:

Step 1: Definition of function of an operator

A function $V(\hat{q})$ of an operator is defined via its power series expansion (assuming V is analytic):

$$V(\hat{q}) = \sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!} \hat{q}^n \tag{73}$$

where $V^{(n)}(0)$ is the n-th derivative of V evaluated at 0.

Step 2: Action of powers of $\hat{q}$ on eigenstates

By definition of the position eigenstate:

$$\hat{q}|q\rangle = q|q\rangle \tag{74}$$

Apply $\hat{q}$ again:

$$\hat{q}^2|q\rangle = \hat{q}(\hat{q}|q\rangle) = \hat{q}(q|q\rangle) = q(\hat{q}|q\rangle) = q^2|q\rangle \tag{75}$$

By induction, for any integer $n \geq 0$:

$$\hat{q}^n|q\rangle = q^n|q\rangle \tag{76}$$

Step 3: Action of $V(\hat{q})$ on eigenstates

Apply the power series to $|q\rangle$:

$$V(\hat{q})|q\rangle = \left(\sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!}\hat{q}^n\right)|q\rangle \tag{77}$$

$$= \sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!}\hat{q}^n|q\rangle \tag{78}$$

$$= \sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!}q^n|q\rangle \tag{79}$$

$$= \left(\sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!}q^n\right)|q\rangle \tag{80}$$

$$= V(q)|q\rangle \tag{81}$$

where the last line recognizes the power series for $V(q)$.

Alternative proof (spectral theorem):

For any Hermitian operator $\hat{q}$ with continuous spectrum, the spectral theorem gives:

$$\hat{q} = \int dq\, q\, |q\rangle\langle q| \tag{82}$$

A function of the operator is:

$$V(\hat{q}) = \int dq\, V(q)\, |q\rangle\langle q| \tag{83}$$

Apply to a position eigenstate $|q_0\rangle$:

$$V(\hat{q})|q_0\rangle = \int dq\, V(q)\, |q\rangle\langle q|q_0\rangle \tag{84}$$

$$= \int dq\, V(q)\, |q\rangle\delta(q - q_0) \tag{85}$$

$$= V(q_0)|q_0\rangle \tag{86}$$

This proves the lemma. □

Step 4: Exponential of $V(\hat{q})$

Since $e^{-\epsilon V(\hat{q})}$ is a function of $\hat{q}$ (with $f(x) = e^{-\epsilon V(x)}$), the lemma applies:

$$e^{-\epsilon V(\hat{q})}|q_k\rangle = e^{-\epsilon V(q_k)}|q_k\rangle \tag{87}$$

Step 5: Matrix element

Now take the inner product with $\langle p_k|$:

$$\langle p_k|e^{-\epsilon V(\hat{q})}|q_k\rangle = e^{-\epsilon V(q_k)}\langle p_k|q_k\rangle \tag{88}$$

Using the position-momentum overlap $\langle p|q\rangle = e^{-ipq}$ (derived in Appendix A):

$$\boxed{\langle p_k|e^{-\epsilon V(\hat{q})}|q_k\rangle = e^{-\epsilon V(q_k)}e^{-ip_k q_k}} \tag{89}$$

Second factor: $\langle q_{k+1}|e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle$
Since $\hat{p}^2$ acts diagonally on momentum eigenstates:

$$\hat{p}^2|p_k\rangle = p_k^2|p_k\rangle \tag{90}$$

Therefore:

$$e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle = e^{-\epsilon p_k^2/(2m)}|p_k\rangle \tag{91}$$

Now take the inner product with $\langle q_{k+1}|$:

$$\langle q_{k+1}|e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle = e^{-\epsilon p_k^2/(2m)}\langle q_{k+1}|p_k\rangle \tag{92}$$

Using $\langle q|p\rangle = e^{ipq}$:

$$\langle q_{k+1}|e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle = e^{-\epsilon p_k^2/(2m)}e^{ip_k q_{k+1}} \tag{93}$$

Combine:

$$\langle q_{k+1}|e^{-\epsilon H}|q_k\rangle = \int \frac{dp_k}{2\pi}\exp\left[-\frac{\epsilon p_k^2}{2m} + ip_k(q_{k+1}-q_k) - \epsilon V(q_k)\right] \tag{94}$$

3.5.3 Gaussian Momentum Integral

Complete the square:

$$\int \frac{dp}{2\pi}\exp\left[-\frac{\epsilon p^2}{2m} + ip\Delta q\right] = \sqrt{\frac{m}{2\pi\epsilon}}\exp\left[-\frac{m(\Delta q)^2}{2\epsilon}\right] \tag{95}$$

Result:

$$\langle q_{k+1}|e^{-\epsilon H}|q_k\rangle = \sqrt{\frac{m}{2\pi\epsilon}}\exp\left[-\epsilon\frac{m}{2}\left(\frac{q_{k+1}-q_k}{\epsilon}\right)^2 - \epsilon V(q_k)\right] \tag{96}$$

3.5.4 Continuum Limit

Take $N \to \infty$, $\epsilon \to 0$:

$$\frac{q_{k+1}-q_k}{\epsilon} \to \frac{dq}{d\tau} \equiv \dot{q}(\tau) \tag{97}$$

$$\epsilon\sum_k \to \int_0^\beta d\tau \tag{98}$$

$$\prod_k dq_k\sqrt{\frac{m}{2\pi\epsilon}} \to \mathcal{D}q \text{ (functional measure)} \tag{99}$$

Final path integral:

$$\langle q_f|e^{-\beta H}|q_i\rangle = \int_{q(0)=q_i}^{q(\beta)=q_f}\mathcal{D}q\, e^{-S_E[q]} \tag{100}$$

where Euclidean action:

$$S_E[q] = \int_0^\beta d\tau\left[\frac{1}{2}m\dot{q}^2 + V(q)\right] \tag{101}$$

3.6 Path Integral Derivation: Quantum Field Theory

3.6.1 Lattice Discretization

Space: Discretize into M sites with spacing a:

$$x_j = j \cdot a \quad \text{for } j = 0, 1, \ldots, M-1 \tag{102}$$

Field at site j: $\phi_j(\tau) \equiv \phi(x_j, \tau)$
Conjugate momentum: $\pi_j(\tau)$
Hamiltonian:

$$H = \sum_{j=0}^{M-1} a \left[\frac{\pi_j^2}{2} + \frac{(\partial_x \phi_j)^2}{2} + V(\phi_j) \right] \tag{103}$$

where $(\partial_x \phi_j) \approx (\phi_{j+1} - \phi_j)/a$.

3.6.2 Field Eigenstates

Define:

$$|\phi\rangle \equiv |\{\phi_j\}\rangle = |\phi_0\rangle \otimes |\phi_1\rangle \otimes \cdots \otimes |\phi_{M-1}\rangle \tag{104}$$

Eigenvalue equation:

$$\hat{\phi}_j |\phi_k\rangle = \phi_j \delta_{jk} |\phi_k\rangle \tag{105}$$

Orthogonality:

$$\langle \phi | \phi' \rangle = \prod_{j=0}^{M-1} \delta(\phi_j - \phi'_j) \tag{106}$$

Completeness:

$$\int \mathcal{D}\phi \, |\phi\rangle\langle\phi| = I, \quad \mathcal{D}\phi = \prod_{j=0}^{M-1} d\phi_j \tag{107}$$

Clarification: Complex coordinate z in 2D CFT **Important:** In 2D conformal field theory, the complex coordinate z represents a point in 2D Euclidean spacetime, NOT a 3D spatial point.

$$z = x + i\tau, \qquad \bar{z} = x - i\tau \tag{108}$$

where:

- x = spatial coordinate (length)
- τ = Euclidean time coordinate (length, after setting $c = 1$)

Dimensions of parameters in the uniformizing map:
The uniformizing map:

$$w(z) = \left(\frac{z-u}{z-v} \right)^{1/n} \tag{109}$$

Symbol	Physical Meaning	Dimension
z, w	Complex coordinates on $\mathcal{R}_n$ and $\mathbb{C}$	Length
u, v	Endpoints of interval $A = [u, v]$	Length
n	Replica number (integer, analytically continued)	Dimensionless
c	Central charge (counts degrees of freedom)	Dimensionless
Δ_n	Scaling dimension of twist field	Dimensionless
$T(z)$	Holomorphic stress tensor	Length^{-2}
$\{w, z\}$	Schwarzian derivative	Length^{-2}
ϵ	UV cutoff (lattice spacing)	Length

- Both $z - u$ and $z - v$ have dimensions of length
- The ratio $\frac{z-u}{z-v}$ is dimensionless
- Raising to power $1/n$ (dimensionless) gives a dimensionless number
- In practice, we work in units where the UV cutoff $\epsilon = 1$, or absorb the scale into the definition of w

Why z is 2D, not 3D: In higher-dimensional QFT (e.g., 3+1 dimensions), the entangling surface is typically planar or spherical. The replica trick introduces a conical singularity in the 2D plane *transverse* to the entangling surface. The uniformizing map applies to the complex coordinate $z = re^{i\theta}$ in that transverse 2D plane. Thus, z remains a 2D complex coordinate even when the full theory lives in higher dimensions.

Dimensional consistency: The Schwarzian derivative $\{w, z\}$ has dimension Length^{-2}, matching the stress tensor $T(z)$. The central charge c is dimensionless, ensuring the transformation law

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\} \tag{110}$$

is dimensionally consistent.

3.6.3 Continuum Limit

Take $a \to 0$:

$$\sum_j a \to \int dx \tag{111}$$

$$\phi_j(\tau) \to \phi(x, \tau) \tag{112}$$

$$\pi_j(\tau) \to \pi(x, \tau) \tag{113}$$

Euclidean action:

$$S_E[\phi] = \int_0^\beta d\tau \int dx \left[\frac{1}{2}(\partial_\tau \phi)^2 + \frac{1}{2}(\partial_x \phi)^2 + V(\phi)\right] \tag{114}$$

Path integral:

$$\langle\phi_f|e^{-\beta H}|\phi_i\rangle = \int_{\phi(x,0)=\phi_i(x)}^{\phi(x,\beta)=\phi_f(x)} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{115}$$

3.7 Thermal Density Matrix

Thermal density operator:

$$\hat{\rho} = \frac{e^{-\beta H}}{Z}, \quad Z = \mathrm{Tr}\big(e^{-\beta H}\big) \tag{116}$$

Matrix elements in field basis:

$$\rho(\phi_f|\phi_i) = \langle\phi_f|\hat{\rho}|\phi_i\rangle = \frac{1}{Z}\langle\phi_f|e^{-\beta H}|\phi_i\rangle \tag{117}$$

Path integral representation:

$$\rho(\phi_f|\phi_i) = \frac{1}{Z}\int_{\phi(x,0)=\phi_i(x)}^{\phi(x,\beta)=\phi_f(x)} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{118}$$

Delta function form (Eq. 7 of Calabrese-Cardy):

$$\rho(\phi_f|\phi_i) = \frac{1}{Z}\int \mathcal{D}\phi\, e^{-S_E[\phi]} \prod_x \delta(\phi(x,0)-\phi_i(x))\delta(\phi(x,\beta)-\phi_f(x)) \tag{119}$$

Interpretation: The delta function product enforces boundary conditions on the path integral:

- $\prod_x \delta(\phi(x,0)-\phi_i(x))$: fixes the *initial condition* at $\tau = 0$ to be $\phi_i(x)$
- $\prod_x \delta(\phi(x,\beta)-\phi_f(x))$: fixes the *final condition* at $\tau = \beta$ to be $\phi_f(x)$
- The functional integral $\int \mathcal{D}\phi$ is now over *all* field configurations without boundary restrictions
- The delta functions select only those configurations satisfying the boundary conditions

Equivalence: This form is equivalent to the boundary-condition notation:

$$\int_{\phi(x,0)=\phi_i(x)}^{\phi(x,\beta)=\phi_f(x)} \mathcal{D}\phi\, e^{-S_E[\phi]} = \int \mathcal{D}\phi\, e^{-S_E[\phi]} \prod_x \delta(\phi(x,0)-\phi_i(x))\delta(\phi(x,\beta)-\phi_f(x)) \tag{120}$$

The delta function form is useful when computing traces or partial traces, as it makes the boundary conditions explicit in the integrand.

Partition function as trace:

$$Z = \mathrm{Tr}\big(e^{-\beta H}\big) = \int \mathcal{D}\phi\, \langle\phi|e^{-\beta H}|\phi\rangle = \int_{\phi(x,0)=\phi(x,\beta)} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{121}$$

Geometry: Cylinder with circumference β.

3.8 Reduced Density Matrix and Partial Trace

Spatial partition:

$$\text{Space} = A \cup B \quad \text{(disjoint)} \tag{122}$$

where A is the subsystem of interest (e.g., interval $[u, v]$) and B is the complement.

Hilbert space factorization:

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B, \quad |\phi\rangle = |\phi_A\rangle \otimes |\phi_B\rangle \tag{123}$$

Reduced density matrix:

$$\rho_A = \mathrm{Tr}_B(\rho) \tag{124}$$

Matrix elements:

$$\langle \phi_A | \rho_A | \phi'_A \rangle = \int \mathcal{D}\phi_B \, \langle \phi_A, \phi_B | \rho | \phi'_A, \phi_B \rangle \tag{125}$$

Path integral derivation:

$$\langle \phi_A | \rho_A | \phi'_A \rangle = \frac{1}{Z} \int \mathcal{D}\phi_B \int \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{126}$$

$$\times \prod_{x \in A} \delta(\phi(x, 0) - \phi'_A(x)) \delta(\phi(x, \beta) - \phi_A(x)) \tag{127}$$

$$\times \prod_{x \in B} \delta(\phi(x, 0) - \phi_B(x)) \delta(\phi(x, \beta) - \phi_B(x)) \tag{128}$$

Key identity:

$$\int d\phi_x \, \delta(\phi(x, 0) - \phi_x) \delta(\phi(x, \beta) - \phi_x) = \delta(\phi(x, 0) - \phi(x, \beta)) \tag{129}$$

Result:

$$\langle \phi_A | \rho_A | \phi'_A \rangle = \frac{1}{Z} \int \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{130}$$

$$\times \prod_{x \in A} \delta(\phi(x, 0) - \phi'_A(x)) \delta(\phi(x, \beta) - \phi_A(x)) \tag{131}$$

$$\times \prod_{x \in B} \delta(\phi(x, 0) - \phi(x, \beta)) \tag{132}$$

Geometric interpretation:

- Region A: Open cut (matrix indices ϕ_A and ϕ'_A)
- Region B: Sewn together (periodic boundary condition)

Boundary conditions summary:
Geometry: Cylinder with cuts along region A.

Region	Boundary Condition	Geometry
A	$\phi(x,0) = \phi'_A$, $\phi(x,\beta) = \phi_A$	Open cut (matrix indices)
B	$\phi(x,0) = \phi(x,\beta)$	Sewn together (periodic)

3.9 Comparison: Discrete vs. Continuum Partial Trace

To clarify the path integral derivation, we compare the standard discrete partial trace from quantum mechanics with the functional integral over region B.

3.9.1 Standard Discrete Partial Trace

Consider a bipartite system with Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, where $\mathcal{H}_A$ and $\mathcal{H}_B$ are finite-dimensional. Let $\{|a\rangle\}$ be an orthonormal basis for $\mathcal{H}_A$ and $\{|b\rangle\}$ for $\mathcal{H}_B$. The full basis is $\{|a,b\rangle = |a\rangle \otimes |b\rangle\}$.

A general density matrix ρ can be expanded as:

$$\rho = \sum_{a,b,a',b'} \rho_{ab,a'b'} \, |a,b\rangle\langle a',b'| \tag{133}$$

The reduced density matrix for subsystem A is obtained by tracing over B:

$$\rho_A = \mathrm{Tr}_B(\rho) = \sum_b \langle b|\rho|b\rangle_B \tag{134}$$

where the subscript on the inner product indicates the partial inner product over B only. In terms of matrix elements:

$$(\rho_A)_{aa'} = \langle a|\rho_A|a'\rangle = \sum_b \langle a,b|\rho|a',b\rangle = \sum_b \rho_{ab,a'b'} \tag{D1}$$

Key observation: The partial trace sums over the diagonal elements of subsystem B in any chosen basis. The indices corresponding to B are set equal in the bra and ket before summation.

3.9.2 Path Integral Partial Trace (Continuum Version)

In quantum field theory, the basis states are field configurations ϕ_A on A and ϕ_B on B. The partial trace is:

$$\langle \phi_A|\rho_A|\phi'_A\rangle = \int \mathcal{D}\phi_B \, \langle \phi_A, \phi_B|\rho|\phi'_A, \phi_B\rangle \tag{C1}$$

This is the exact continuous analog of (D1):

3.9.3 From Trace to Sewing: The Key Delta Function Identity

In the discrete case, we can write the sum over b using a Kronecker delta to emphasize the equality of indices:

$$\sum_b \delta_{b,b_1}\delta_{b,b_2} = \delta_{b_1,b_2} \tag{D2}$$

Discrete (D1)	**Continuous (C1)**
Sum over basis index b	Functional integral over field ϕ_B
Matrix element $\rho_{ab,a'b}$	Kernel $\langle \phi_A, \phi_B \vert \rho \vert \phi'_A, \phi_B \rangle$
Summation $\sum_b$	Integration $\int \mathcal{D}\phi_B$

This trivial identity is the discrete analog of the functional identity used in the path integral derivation:

$$\int \mathcal{D}\phi_B \prod_{\mathbf{x}\in B} \delta\big(\phi(\mathbf{x},0) - \phi_B(\mathbf{x})\big)\delta\big(\phi(\mathbf{x},\beta) - \phi_B(\mathbf{x})\big) = \prod_{\mathbf{x}\in B} \delta\big(\phi(\mathbf{x},0) - \phi(\mathbf{x},\beta)\big) \tag{C2}$$

Both equations state: *summing over a shared index forces the two appearances of that index to be equal.*

3.9.4 Geometric Interpretation Comparison

Discrete case:

- We start with a density matrix that has independent bra and ket indices for B: $\rho_{ab,a'b'}$
- The partial trace sets $b = b'$ and sums over b
- Visual: The two "legs" of the matrix for subsystem B are connected (traced together)

Continuum path integral case:

- The thermal density matrix is represented by a path integral on a cylinder of circumference β, with independent boundary conditions at $\tau = 0$ (bra) and $\tau = \beta$ (ket)
- On region A, the bra and ket boundary values ϕ'_A and ϕ_A remain independent (matrix indices)
- On region B, the partial trace forces ϕ_B at $\tau = 0$ and $\tau = \beta$ to be equal, and then integrates over ϕ_B
- This is exactly sewing the two ends of the cylinder together on B, making the time direction periodic there

3.9.5 Lattice Version as Direct Bridge

The discrete lattice formulation makes the correspondence explicit. Let spatial lattice sites be partitioned into A and B. The density matrix is $\rho(\{\phi_i\}, \{\phi'_i\})$.

The discrete partial trace is:

$$\rho_A(\{\phi_A\}, \{\phi'_A\}) = \int \prod_{j\in B} d\phi_j \; \rho\big(\{\phi_A, \phi_B\}, \{\phi'_A, \phi_B\}\big) \tag{L1}$$

When ρ is expressed via a lattice path integral (with time slices $\tau = 0, \ldots, N_\tau$), we get:

- On sites in B: $\sum_{\phi_j} \delta_{\phi(j,0),\phi_j}\delta_{\phi(j,N_\tau),\phi_j} = \delta_{\phi(j,0),\phi(j,N_\tau)}$
- On sites in A: $\delta_{\phi(i,0),\phi'_A(i)}\delta_{\phi(i,N_\tau),\phi_A(i)}$

This is the perfect lattice parallel to the continuous delta functional identity.

3.9.6 Why This Matters for Entanglement Entropy

The reduced density matrix ρ_A is the central object for computing entanglement entropy $S_A = -\operatorname{Tr}(\rho_A \log \rho_A)$. In the replica trick, one computes $\operatorname{Tr}(\rho_A^n)$ by taking n copies of the cut cylinder and cyclically gluing them along the cuts on A.

- The sewing on B comes from the partial trace over B
- The cyclic gluing on A comes from the matrix multiplication in ρ_A^n

Thus, understanding the partial trace as a sewing operation on B is essential for the geometric computation of entanglement entropy in QFT.

3.9.7 Summary Table

Aspect	**Discrete QM Partial Trace**	**QFT Path Integral Partial Trace**
Trace operation	$\sum_b \langle b\|\rho\|b\rangle$	$\int \mathcal{D}\phi_B \langle \phi_A, \phi_B\|\rho\|\phi'_A, \phi_B\rangle$
Index setting	$b = b'$ (bra and ket index equal)	ϕ_B same at $\tau = 0$ and $\tau = \beta$
Delta identity	$\sum_b \delta_{b,b_1}\delta_{b,b_2} = \delta_{b_1,b_2}$	$\int \mathcal{D}\phi_B \delta(\phi(0) - \phi_B)\delta(\phi(\beta) - \phi_B) = \delta(\phi(0) - \phi(\beta))$
Geometric result	Connect subsystem B legs	Sew cylinder on region B (periodic boundary)

The path integral partial trace is the natural functional generalization of the standard discrete partial trace, with the sum over a complete set of states replaced by a functional integral and the index equality enforced by a Dirac delta functional.

3.10 Density Matrix as Path Integral

The thermal density matrix at temperature β^{-1} is:

$$\rho = \frac{e^{-\beta H}}{Z}, \quad Z = \operatorname{Tr}\left(e^{-\beta H}\right) \tag{135}$$

In path integral form:

$$\rho(\{\phi_x\}, \{\phi'_x\}) = \frac{1}{Z}\int \mathcal{D}\phi\, e^{-S_E[\phi]} \prod_x \delta(\phi(x,0) - \phi'_x)\delta(\phi(x,\beta) - \phi_x) \tag{136}$$

where S_E is the Euclidean action.

3.11 Reduced Density Matrix: Tracing Out Region B

To obtain $\rho_A = \mathrm{Tr}_B\, \rho$, we integrate over field configurations in region B:

$$\rho_A(\{\phi_A\}, \{\phi'_A\}) = \int \mathcal{D}\phi_B\, \rho(\{\phi_A, \phi_B\}, \{\phi'_A, \phi_B\}) \tag{137}$$

Geometric interpretation: The path integral is on a cylinder with:

- Region B: sewn together (periodic boundary condition)
- Region A: cut open (boundary conditions ϕ_A and ϕ'_A)

3.12 Replica Construction

The trace $\mathrm{Tr}(\rho_A^n)$ can be expressed as a product of matrix elements.

Step 1: Start with trace definition. From Eq. (15):

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi\, \langle\phi|\rho_A^n|\phi\rangle \tag{138}$$

Step 2: Expand ρ_A^n as operator product:

$$\rho_A^n = \underbrace{\rho_A \cdot \rho_A \cdot \cdots \cdot \rho_A}_{n \text{ factors}} \tag{139}$$

Therefore:

$$\langle\phi|\rho_A^n|\phi\rangle = \langle\phi|\rho_A \cdot \rho_A \cdot \cdots \cdot \rho_A|\phi\rangle \tag{140}$$

Step 3: Insert identity between each factor. Completeness relation:

$$I = \int \mathcal{D}\phi'\, |\phi'\rangle\langle\phi'| \tag{141}$$

For $n = 2$:

$$\langle\phi|\rho_A^2|\phi\rangle = \langle\phi|\rho_A \cdot \rho_A|\phi\rangle \tag{142}$$

$$= \langle\phi|\rho_A \left(\int \mathcal{D}\phi^{(2)}\, |\phi^{(2)}\rangle\langle\phi^{(2)}|\right) \rho_A|\phi\rangle \tag{143}$$

$$= \int \mathcal{D}\phi^{(2)}\, \langle\phi|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi\rangle \tag{144}$$

For $n = 3$:

$$\langle\phi|\rho_A^3|\phi\rangle = \langle\phi|\rho_A \cdot \rho_A \cdot \rho_A|\phi\rangle \tag{145}$$

$$= \int \mathcal{D}\phi^{(2)}\mathcal{D}\phi^{(3)}\, \langle\phi|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle\langle\phi^{(3)}|\rho_A|\phi\rangle \tag{146}$$

For general n:

$$\langle\phi|\rho_A^n|\phi\rangle = \int \mathcal{D}\phi^{(2)} \cdots \mathcal{D}\phi^{(n)}\, \langle\phi|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle \cdots \langle\phi^{(n)}|\rho_A|\phi\rangle \tag{147}$$

Step 4: Relabel for cyclic symmetry. Define $\phi^{(1)} \equiv \phi$:

$$\langle\phi^{(1)}|\rho_A^n|\phi^{(1)}\rangle = \int \mathcal{D}\phi^{(2)} \cdots \mathcal{D}\phi^{(n)}\, \langle\phi^{(1)}|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle \cdots \langle\phi^{(n)}|\rho_A|\phi^{(1)}\rangle \tag{148}$$

3.12.1 Clarification: Replica Indices vs. Time Slices

Important: The superscripts $(1), (2), \ldots, (n)$ label *replicas* (copies of the system), NOT time slices.

Replica Indices	**Time Slices**
$\phi^{(1)}, \phi^{(2)}, \ldots, \phi^{(n)}$	$\tau_1, \tau_2, \ldots, \tau_{N_\tau}$
n independent copies of the *entire* system	One system discretized in Euclidean time
Used for computing $\mathrm{Tr}(\rho_A^n)$ (replica trick)	Used for defining the path integral
Each replica has its own time direction $\tau \in [0, \beta]$	Time slices live within one replica
Cyclic gluing: copy j connects to copy $j+1$ along region A	Time evolution: τ_k connects to τ_{k+1}

Visual distinction:

- **Replicas:** Imagine n separate cylinders (each representing a full path integral from $\tau = 0$ to $\tau = \beta$). We glue them together along the cut in region A to form an n-sheeted Riemann surface.
- **Time slices:** Within *each* cylinder, we discretize the time direction into N_τ slices to define the functional measure $\mathcal{D}\phi = \prod_{k=1}^{N_\tau} d\phi_k$.

Why the confusion arises: Both involve "copies" in some sense:

- Time slices: We copy the field variable at each time step $\phi(\tau_k)$
- Replicas: We copy the *entire system* (all time slices together) n times

Key insight: The replica index (j) labels which *sheet* of the Riemann surface we are on. When we integrate over $\phi^{(j)}$, we are gluing sheet $j-1$ to sheet j along region A. This is completely independent of the time discretization within each sheet.

Example for $n = 3$:

$$\mathrm{Tr}(\rho_A^3) = \int \mathcal{D}\phi^{(1)}\mathcal{D}\phi^{(2)}\mathcal{D}\phi^{(3)} \, \langle\phi^{(1)}|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle\langle\phi^{(3)}|\rho_A|\phi^{(1)}\rangle \tag{149}$$

$$= \int \mathcal{D}\phi^{(1)}\mathcal{D}\phi^{(2)}\mathcal{D}\phi^{(3)} \underbrace{\left(\int_{\phi^{(2)}}^{\phi^{(1)}} \mathcal{D}\phi \, e^{-S_E}\right)}_{\text{Replica 1}} \underbrace{\left(\int_{\phi^{(3)}}^{\phi^{(2)}} \mathcal{D}\phi \, e^{-S_E}\right)}_{\text{Replica 2}} \underbrace{\left(\int_{\phi^{(1)}}^{\phi^{(3)}} \mathcal{D}\phi \, e^{-S_E}\right)}_{\text{Replica 3}} \tag{150}$$

Each replica integral contains its own time slices $\tau_1, \ldots, \tau_{N_\tau}$, but the replica index (j) tells us which copy of the cylinder we are on.

3.12.2 Clarification: What Distinguishes Replica Sheets

Question: If the bulk action $S_E[\phi^{(j)}]$ is identical for all replicas, what distinguishes sheet (1) from sheet (2), etc.?

Answer: The distinction lies entirely in the *boundary conditions* and the *gluing order*, not in the intrinsic properties of the sheets.

Identical intrinsic properties:

- **Action:** $S_E[\phi^{(1)}] = S_E[\phi^{(2)}] = \cdots = S_E[\phi^{(n)}]$ (same functional form)
- **Manifold geometry:** Each replica is a cylinder of circumference β in the τ direction
- **Field content:** Same fields, same Lagrangian, same coupling constants
- **Bulk physics:** Identical on all sheets

What differs:

- **Boundary values on A:** Each replica j has different boundary conditions $(\phi_A^{(j+1)}, \phi_A^{(j)})$ at $\tau = 0$ and $\tau = \beta$
- **Position in the cyclic chain:** Replica j connects to replica $j+1$ along region A
- **Role in the trace:** The matrix element $\langle \phi^{(j)} | \rho_A | \phi^{(j+1)} \rangle$ appears at position j in the product

Identical for All Replicas	**Differs Across Replicas**
Bulk action $S_E[\phi]$	Boundary values $(\phi_A^{(j+1)}, \phi_A^{(j)})$
Cylinder geometry	Position in cyclic gluing chain
Field content and Lagrangian	Role in matrix product
Central charge c	Integration variable in $\mathrm{Tr}(\rho_A^n)$

Key insight: The replicas are *mathematical copies* introduced to compute $\mathrm{Tr}(\rho_A^n)$. They are not physically distinct systems. The only distinction is their position in the cyclic chain, which determines how they are glued together along region A. After gluing, the n sheets become a single connected manifold $\mathcal{R}_n$, and the replica index (j) simply labels which sheet of the Riemann surface we are on.

Step 5: Substitute back into trace:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi^{(1)} \, \langle \phi^{(1)} | \rho_A^n | \phi^{(1)} \rangle \tag{151}$$

$$= \int \mathcal{D}\phi^{(1)} \mathcal{D}\phi^{(2)} \cdots \mathcal{D}\phi^{(n)} \, \langle \phi^{(1)} | \rho_A | \phi^{(2)} \rangle \langle \phi^{(2)} | \rho_A | \phi^{(3)} \rangle \cdots \langle \phi^{(n)} | \rho_A | \phi^{(1)} \rangle \tag{152}$$

Final result:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi^{(1)} \cdots \mathcal{D}\phi^{(n)} \, \langle \phi^{(1)} | \rho_A | \phi^{(2)} \rangle \langle \phi^{(2)} | \rho_A | \phi^{(3)} \rangle \cdots \langle \phi^{(n)} | \rho_A | \phi^{(1)} \rangle \tag{153}$$

Geometric picture:

- Each matrix element $\langle \phi^{(j)} | \rho_A | \phi^{(j+1)} \rangle$ represents one copy of the system
- The cyclic structure connects copy j to copy $j+1$
- Integration over all $\phi^{(k)}$ sews the copies together
- Result: n-sheeted Riemann surface $\mathcal{R}_n$

3.12.3 What Distinguishes the Replica Sheets?

Question: If each replica is a copy of the same system, what makes sheet (1) different from sheet (2)?

Answer: The sheets are *mathematically identical* but *distinguished by their gluing order*.

Identical intrinsic properties:

- Same Euclidean action: $S_E[\phi^{(1)}] = S_E[\phi^{(2)}] = \cdots = S_E[\phi^{(n)}]$
- Same spacetime geometry: cylinder of circumference β
- Same boundary conditions on region B: periodic in imaginary time
- Same open cut on region A: independent bra and ket field values

Distinction via gluing pattern: The difference emerges from the *cyclic identification* on region A:

- Sheet j at $\tau = \beta$ is glued to sheet $j+1$ at $\tau = 0$
- Cyclic condition: sheet n glues back to sheet 1
- Gluing enforced by delta functions:

$$\prod_{j=1}^{n} \delta\big(\phi^{(j)}(\mathbf{x}, \beta)|_A - \phi^{(j+1)}(\mathbf{x}, 0)|_A\big), \quad (n+1 \equiv 1) \tag{154}$$

Field variable independence: Before gluing, each sheet has independent bulk fields:

$$S_{\text{total}} = \sum_{j=1}^{n} S_E[\phi^{(j)}] \tag{155}$$

The only coupling comes from the boundary delta functions on region A.

Geometric interpretation: After gluing, the n sheets form a single connected manifold $\mathcal{R}_n$. The replica index becomes a *sheet label* on a branched covering:

- Moving around the entangling surface (boundary of A), the sheet index increases by one
- This is a *topological* distinction: locally identical sheets, globally arranged in a cyclic stack
- The branch cut lies along region A

Key insight: Replica sheets are identical copies that differ only in their prescribed gluing order. The replica index (j) is a bookkeeping label that tells us which sheet connects to which in the cyclic trace.

Within one sheet	**Across sheets**
Same action S_E	Summed: $\sum_j S_E[\phi^{(j)}]$
Independent bulk field $\phi^{(j)}$	Glued on A: $\phi^{(j)}(\beta) = \phi^{(j+1)}(0)$
Identical cylinder geometry	Cyclic stack with branch cut along A
No intrinsic replica label	Label j distinguishes gluing order

Table 1: Comparison of properties within and across replica sheets.

Bulk action is identical: Yes, the Euclidean action $S_E[\phi]$ is exactly the same for every replica copy. The copies are dynamically identical; the difference is only in how they are glued together at the boundaries.

Bulk action per replica:

$$S_E[\phi^{(j)}] = \int d^d x \int_0^\beta d\tau\, \mathcal{L}_E[\phi^{(j)}, \partial\phi^{(j)}] \tag{156}$$

There is no difference in the Lagrangian, coupling constants, or geometry within each replica. The copies are true identical replicas.

Total action in the product:

$$S_{\text{total}} = \sum_{j=1}^{n} S_E[\phi^{(j)}] \tag{157}$$

The copies do not interact directly through the bulk action — they are coupled *only* by the cyclic boundary conditions on region A.

Where the distinction lies:

- **Bulk action:** Identical S_E for all j
- **Boundary values on A:** Different for each copy
  - Copy j: ($\phi_A^{(j+1)}$ at $\tau = 0$, $\phi_A^{(j)}$ at $\tau = \beta$)
  - Copy $j+1$: ($\phi_A^{(j+2)}$ at $\tau = 0$, $\phi_A^{(j+1)}$ at $\tau = \beta$)
- **Role in gluing:** Different position in the cyclic chain

Analogy: Quantum mechanics matrix elements

Consider a single-particle density matrix $\rho(x, x') = \langle x|e^{-\beta H}|x'\rangle/Z_1$. The Hamiltonian H is the same regardless of whether we compute $\rho(x_1, x_2)$ or $\rho(x_2, x_3)$. The only difference is the endpoints in the path integral. The replica method is the functional field theory analog.

After gluing: Single action on branched manifold

Once we integrate over the intermediate $\phi_A^{(j)}$ and $\phi_B^{(j)}$, the delta functions glue the replicas together. The resulting object is a single connected manifold $\mathcal{R}_n$ (the n-sheeted surface). The path integral on $\mathcal{R}_n$ can be written as:

$$Z_n = \int_{\mathcal{R}_n} \mathcal{D}\Phi\, e^{-S_E[\Phi]} \tag{158}$$

where Φ is a single field defined on the entire branched cover, and $S_E[\Phi]$ is still the same local action, now integrated over the full surface. The fact that the surface has a branch cut along A does not change the Lagrangian; it only changes the global topology over which the action is evaluated.

Feature	**Comparison**
Bulk action	Identical S_E for all copies
Manifold geometry	Same cylinder for each copy
Boundary values on A	Different: $(\phi_A^{(j+1)}, \phi_A^{(j)})$ vs $(\phi_A^{(j+2)}, \phi_A^{(j+1)})$
Role in gluing	Different position in cycle
After gluing	Single field Φ on $\mathcal{R}_n$

Table 2: Action and boundary conditions across replica copies.

Summary: The actions are identical, but the boundary conditions (and therefore the specific field configurations summed over) differ for each copy in the product. This is precisely what allows the trace to be non-trivial: the matrix multiplication forces the path integrals to communicate via the shared boundary values.

3.13 Partition Function Ratio

The key result is:

$$\mathrm{Tr}(\rho_A^n) = \frac{Z_n}{(Z_1)^n} \tag{159}$$

where:

- Z_n: partition function on the n-sheeted surface $\mathcal{R}_n$
- Z_1: partition function on a single copy (normalized to 1)

3.13.1 Derivation of the Ratio

Step 1: Definition of partition functions

The partition function on a manifold $\mathcal{M}$ is:

$$Z[\mathcal{M}] = \int_{\mathcal{M}} \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{160}$$

For the replica trick:

- $Z_1 = Z[\mathbb{C}]$: partition function on the complex plane (single sheet)
- $Z_n = Z[\mathcal{R}_n]$: partition function on the n-sheeted Riemann surface

Step 2: Normalization of the density matrix

The thermal density matrix is normalized:

$$\mathrm{Tr}(\rho) = 1 \quad \Rightarrow \quad \frac{1}{Z_1} \int \mathcal{D}\phi \, e^{-S_E[\phi]} = 1 \tag{161}$$

Therefore, the normalized density matrix element is:

$$\rho(\phi_f|\phi_i) = \frac{1}{Z_1}\int_{\phi_i}^{\phi_f} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{162}$$

Step 3: Product of density matrices
From the replica construction (Eq. 59):

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi^{(1)}\cdots\mathcal{D}\phi^{(n)}\, \rho_A(\phi^{(1)}|\phi^{(2)})\rho_A(\phi^{(2)}|\phi^{(3)})\cdots\rho_A(\phi^{(n)}|\phi^{(1)}) \tag{163}$$

Substitute the path integral form:

$$\rho_A(\phi^{(j)}|\phi^{(j+1)}) = \frac{1}{Z_1}\int_{\phi^{(j+1)}}^{\phi^{(j)}} \mathcal{D}\phi^{(j)}\, e^{-S_E[\phi^{(j)}]} \tag{164}$$

where the path integral is over fields on copy j with boundary conditions matching region A.

Step 4: Combine the path integrals
Multiplying n factors:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi^{(1)}\cdots\mathcal{D}\phi^{(n)} \prod_{j=1}^{n}\left[\frac{1}{Z_1}\int_{\phi^{(j+1)}}^{\phi^{(j)}} \mathcal{D}\phi^{(j)}\, e^{-S_E[\phi^{(j)}]}\right] \tag{165}$$

$$= \frac{1}{(Z_1)^n}\int \mathcal{D}\phi^{(1)}\cdots\mathcal{D}\phi^{(n)}\, e^{-\sum_{j=1}^{n} S_E[\phi^{(j)}]} \tag{166}$$

Step 5: Cyclic gluing
The cyclic boundary condition $\phi^{(n+1)} \equiv \phi^{(1)}$ means:

- Copy 1 connects to copy 2 along region A
- Copy 2 connects to copy 3 along region A
- $\cdots$
- Copy n connects back to copy 1 along region A

This creates the n-sheeted Riemann surface $\mathcal{R}_n$.

3.13.2 Detailed Gluing Procedure: From Eq. (154) to $Z_n/(Z_1)^n$

The transition from Equation (154) to the final ratio is precisely the gluing step that converts n independent path integrals into a single path integral on a connected, multi-sheeted manifold. Let's walk through the missing details carefully.

Equation (154) restated with constraints Equation (154) from the notes reads:

$$\mathrm{Tr}(\rho_A^n) = \frac{1}{(Z_1)^n} \int \mathcal{D}\phi^{(1)} \cdots \mathcal{D}\phi^{(n)}\, e^{-\sum_{j=1}^n S_E[\phi^{(j)}]}. \tag{154}$$

But this is incomplete as written. The integrals over $\phi^{(j)}$ are constrained by delta functionals that enforce the boundary conditions from each matrix element $\langle \phi_A^{(j)} | \rho_A | \phi_A^{(j+1)} \rangle$. The full expression (including those constraints) is:

$$\mathrm{Tr}(\rho_A^n) = \frac{1}{(Z_1)^n} \int \left[\prod_{j=1}^n \mathcal{D}\phi^{(j)}\right] \left[\prod_{j=1}^n \mathcal{D}\phi_A^{(j)}\right] \left[\prod_{j=1}^n \mathcal{D}\phi_B^{(j)}\right] e^{-\sum_{j=1}^n S_E[\phi^{(j)}]} \times (\text{delta functionals}). \tag{154-full}$$

The delta functionals are:

- **For each j and each $\mathbf{x} \in A$:**

$$\delta\big(\phi^{(j)}(\mathbf{x},0) - \phi_A^{(j+1)}(\mathbf{x})\big)\, \delta\big(\phi^{(j)}(\mathbf{x},\beta) - \phi_A^{(j)}(\mathbf{x})\big) \tag{167}$$

- **For each j and each $\mathbf{x} \in B$:**

$$\delta\big(\phi^{(j)}(\mathbf{x},0) - \phi_B^{(j)}(\mathbf{x})\big)\, \delta\big(\phi^{(j)}(\mathbf{x},\beta) - \phi_B^{(j)}(\mathbf{x})\big) \tag{168}$$

Step 1: Integrating over $\phi_B^{(j)}$ (Region B) For each replica j, we have an independent functional integral over $\phi_B^{(j)}$. The relevant part of the integrand is:

$$\int \mathcal{D}\phi_B^{(j)} \prod_{\mathbf{x}\in B} \delta\big(\phi^{(j)}(\mathbf{x},0) - \phi_B^{(j)}(\mathbf{x})\big)\, \delta\big(\phi^{(j)}(\mathbf{x},\beta) - \phi_B^{(j)}(\mathbf{x})\big). \tag{169}$$

Using the key delta identity:

$$\int d\psi\, \delta(a-\psi)\delta(b-\psi) = \delta(a-b), \tag{170}$$

this yields:

$$\prod_{\mathbf{x}\in B} \delta\big(\phi^{(j)}(\mathbf{x},0) - \phi^{(j)}(\mathbf{x},\beta)\big). \tag{171}$$

Thus, after integrating out all $\phi_B^{(j)}$, each replica acquires **periodic boundary conditions on region B**:

$$\phi^{(j)}(\mathbf{x},0)|_B = \phi^{(j)}(\mathbf{x},\beta)|_B. \tag{172}$$

Step 2: Integrating over $\phi_A^{(j)}$ (Region A) Now examine the integrals over $\phi_A^{(j)}$. A given replica index k appears in two sets of delta functions:

- From replica $k-1$ (with $0 \equiv n$ for $k=1$):

$$\delta\big(\phi^{(k-1)}(\mathbf{x},\beta) - \phi_A^{(k)}(\mathbf{x})\big) \tag{173}$$

- From replica k:

$$\delta\big(\phi^{(k)}(\mathbf{x},0) - \phi_A^{(k)}(\mathbf{x})\big) \tag{174}$$

The integral over $\phi_A^{(k)}$ is therefore:

$$\int \mathcal{D}\phi_A^{(k)} \prod_{\mathbf{x}\in A} \delta\big(\phi^{(k-1)}(\mathbf{x},\beta) - \phi_A^{(k)}(\mathbf{x})\big)\, \delta\big(\phi^{(k)}(\mathbf{x},0) - \phi_A^{(k)}(\mathbf{x})\big) \tag{175}$$

$$= \prod_{\mathbf{x}\in A} \delta\big(\phi^{(k-1)}(\mathbf{x},\beta) - \phi^{(k)}(\mathbf{x},0)\big). \tag{176}$$

This **cyclically glues** the replicas together along region A:

$$\phi^{(1)}(\beta)|_A = \phi^{(2)}(0)|_A, \quad \phi^{(2)}(\beta)|_A = \phi^{(3)}(0)|_A, \quad \ldots, \quad \phi^{(n)}(\beta)|_A = \phi^{(1)}(0)|_A. \tag{177}$$

Step 3: The resulting manifold $\mathcal{R}_n$ After these integrations, we are left with a single path integral over the n fields $\phi^{(1)}, \ldots, \phi^{(n)}$ that are no longer independent. They satisfy the boundary conditions:

- **Region B:** each $\phi^{(j)}$ is periodic in its own τ direction.
- **Region A:** the fields are linked cyclically across replicas.

These conditions define an n-sheeted Riemann surface $\mathcal{R}_n$:

- Take n copies of the cylinder.
- Cut each copy along region A.
- Glue the $\tau = \beta$ edge of copy j to the $\tau = 0$ edge of copy $j+1$ on A.

The resulting space is a branched cover of the original cylinder, with a branch cut of order n along the boundary of A.

Step 4: Unified field on $\mathcal{R}_n$ Because the replicas are glued together, the n fields $\phi^{(1)}, \ldots, \phi^{(n)}$ can be viewed as a single field Φ defined on the entire surface $\mathcal{R}_n$. The point $(\mathbf{x}, \tau)$ on sheet j is simply a different coordinate patch for the same global manifold. The sum of actions becomes the action of that single field:

$$\sum_{j=1}^{n} S_E[\phi^{(j)}] = S_E[\Phi] \quad \text{on } \mathcal{R}_n. \tag{178}$$

The functional measure $\prod_{j=1}^{n} \mathcal{D}\phi^{(j)}$ with the gluing constraints is exactly the path integral measure on $\mathcal{R}_n$.

Therefore, the combined integral is precisely the partition function on the branched manifold:

$$\int_{\mathcal{R}_n} \mathcal{D}\Phi\, e^{-S_E[\Phi]} = Z_n. \tag{179}$$

Step E: Final ratio Putting the normalization back in, we obtain the replica trick formula:

$$\boxed{\mathrm{Tr}(\rho_A^n) = \frac{Z_n}{(Z_1)^n}}. \tag{180}$$

Step	What Happens
(154) with constraints	Product of n path integrals + delta functions
Integrate over $\phi_B^{(j)}$	Periodic BC on B for each replica
Integrate over $\phi_A^{(j)}$	Cyclic gluing on A across replicas
Identify the glued manifold	n-sheeted Riemann surface $\mathcal{R}_n$
Combine fields and actions	Single field theory on $\mathcal{R}_n$
Recognize the path integral	Z_n, the partition function on $\mathcal{R}_n$

Table 3: Summary of the transition from Eq. (154) to the final result $Z_n/(Z_1)^n$.

The magic of the replica trick: A complicated trace of an operator becomes a clean geometric partition function ratio.

Step 6: Identify Z_n

The combined path integral is precisely the partition function on $\mathcal{R}_n$:

$$\int_{\mathcal{R}_n} \mathcal{D}\phi\, e^{-S_E[\phi]} = Z_n \tag{181}$$

Final result:

$$\boxed{\mathrm{Tr}(\rho_A^n) = \frac{Z_n}{(Z_1)^n}} \tag{182}$$

Physical interpretation:

- Numerator Z_n: path integral on n-sheeted surface (replica manifold)
- Denominator $(Z_1)^n$: normalization from n copies of the density matrix
- The ratio is finite and well-defined even though individual path integrals diverge

4 Twist Fields

Motivation: The twist field approach provides an alternative derivation of the entanglement entropy using conformal symmetry. Instead of working directly with the path integral on $\mathcal{R}_n$, we map the n-sheeted surface to the complex plane and study the transformation properties of the stress tensor.

4.1 The Uniformizing Map: Derivation and Construction

Goal: Find a holomorphic map $w : \mathcal{R}_n \to \mathbb{C}$ that unwraps the n-sheeted Riemann surface onto a single complex plane.

4.1.1 Geometric Problem

We have an n-sheeted Riemann surface $\mathcal{R}_n$ constructed as follows:

- Start with n identical copies of the complex plane (or cylinder)
- Cut each copy along the interval $A = [u, v]$ on the real axis
- Cyclically glue the upper edge of the cut on copy j to the lower edge of the cut on copy $j + 1$ (with copy $n + 1 \equiv 1$)

The resulting surface is a branched cover of the original complex plane, with branch points at $z = u$ and $z = v$. Near these points, the surface looks like a cone with deficit angle $2\pi(1 - 1/n)$.

4.1.2 Desired Properties of the Map

For $w(z)$ to uniformize $\mathcal{R}_n$, it must:

1. **Be single-valued on $\mathcal{R}_n$:** As we move from sheet to sheet across the branch cut, $w(z)$ should change smoothly and return to its original value after n sheets
2. **Have branch points at u and v:** Near these points, w should behave like $w \sim (z - u)^{1/n}$ and $w \sim (z - v)^{-1/n}$ (up to constants), so that the n sheets are unwrapped into a single plane
3. **Map the entire $\mathcal{R}_n$ bijectively to $\mathbb{C}$:** The map should send the two branch points to 0 and ∞, and the rest of the surface to $\mathbb{C} \setminus \{0, \infty\}$

4.1.3 Step 1: Möbius Transformation to Simplify Branch Cut

First, apply a Möbius transformation that sends the branch points u and v to 0 and ∞:

$$\zeta(z) = \frac{z - u}{z - v} \tag{183}$$

This transformation is biholomorphic on each sheet, and it maps:

- $z = u \mapsto \zeta = 0$
- $z = v \mapsto \zeta = \infty$
- The interval $[u, v]$ on the real axis is mapped to the negative real axis $(-\infty, 0]$ (taking the principal branch)

The branch cut now runs from 0 to ∞ along the negative real axis.

4.1.4 Step 2: Power Map to Unwrap the Multi-Sheeted Structure

On $\mathcal{R}_n$, as we circle $z = u$ (or $\zeta = 0$), the argument of ζ increases by $2\pi n$ to return to the same point on the surface. A function like $\zeta^{1/n}$ will be single-valued on $\mathcal{R}_n$ because multiplying the argument by $1/n$ exactly compensates the n-fold covering.

Define:

$$w = \zeta^{1/n} = \left(\frac{z-u}{z-v}\right)^{1/n} \tag{184}$$

Behavior near branch points:

- **Near $z = u$ ($\zeta \approx 0$):** $w \sim (z-u)^{1/n}$. As we go around u on $\mathcal{R}_n$, $z - u$ accumulates a phase $2\pi n$, so w accumulates a phase $2\pi n/n = 2\pi$, returning to its original value. Thus w is single-valued and has a branch point at u.
- **Near $z = v$ ($\zeta \approx \infty$):** Write $w = \zeta^{1/n}$. For large ζ, consider $\tilde{w} = 1/w = \zeta^{-1/n} = \left(\frac{z-v}{z-u}\right)^{1/n}$. Near $z = v$, $\tilde{w} \sim (z-v)^{1/n}$, so w has a pole of order $1/n$. Going around v on $\mathcal{R}_n$ changes the phase by $2\pi n$, and $\tilde{w}$ changes by 2π, so w is single-valued.

4.1.5 Step 3: Global Bijection

On the original n-sheeted surface, the map w takes each of the n sheets and maps them onto different sectors of the w-plane. Specifically:

- The n sheets correspond to the n branches of the $1/n$-th power
- As we cycle through the sheets, the argument of w increases by $2\pi/n$ each time
- The full surface $\mathcal{R}_n$ is mapped one-to-one onto the entire w-plane
- The branch point $z = u$ maps to $w = 0$
- The branch point $z = v$ maps to $w = \infty$
- The branch cut (the glued edges) becomes the positive real axis in the w-plane, where the argument jumps from 0 to 2π across the cut (this is a coordinate singularity, not a physical cut on $\mathcal{R}_n$)

4.1.6 Why "Uniformizing Map"?

The term "uniformizing" refers to the process of converting a multi-valued function (or multi-sheeted surface) into a single-valued function on a simpler domain. The map $w(z)$:

- Takes the complicated n-sheeted geometry $\mathcal{R}_n$
- Maps it to the simple complex plane $\mathbb{C}$
- Makes all functions single-valued when expressed in terms of w

This is analogous to the uniformization theorem in complex analysis, which states that every simply-connected Riemann surface is conformally equivalent to either the complex plane, the unit disk, or the Riemann sphere.

Original Surface $\mathcal{R}_n$	**Uniformized Plane $\mathbb{C}$**
n sheets glued cyclically along $[u, v]$	Single complex plane
Branch points at u and v	Points 0 and ∞
Multi-valued functions of z	Single-valued functions of w
Complicated geometry	Simple geometry

Table 4: Effect of the uniformizing map $w = \left(\frac{z-u}{z-v}\right)^{1/n}$.

4.2 Stress Tensor Transformation

where:

- z: coordinate on the n-sheeted surface $\mathcal{R}_n$
- w: coordinate on the complex plane $\mathbb{C}$

Properties:

- Branch points at $z = u$ and $z = v$ map to $w = 0$ and $w = \infty$
- The n sheets are unwrapped into the full w-plane
- The map is n-to-1 except at branch points

4.3 The Stress Tensor in Conformal Field Theory

What is the stress tensor? In quantum field theory, the stress-energy tensor (or energy-momentum tensor) $T^{\mu\nu}$ encodes the flux of energy and momentum. It appears in the replica trick because it governs how the theory responds to changes in geometry.

4.3.1 Classical Stress Tensor

In classical field theory, for a scalar field ϕ with action $S = \int d^d x\, \mathcal{L}$, the stress tensor is defined via Noether's theorem for spacetime translations:

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L} \tag{185}$$

Physical interpretation:

- T^{00}: energy density
- T^{0i}: momentum density (energy flux)
- T^{ij}: stress (momentum flux)

In a conformally invariant theory, $T^{\mu\nu}$ is traceless: $T^\mu_\mu = 0$.

4.3.2 Stress Tensor in 2D CFT

In two Euclidean dimensions, it is convenient to use complex coordinates:

$$z = x + i\tau, \qquad \bar{z} = x - i\tau \tag{186}$$

The stress tensor then has components:

- **Holomorphic part:** $T(z) \equiv T_{zz}$ (depends only on z)
- **Antiholomorphic part:** $\bar{T}(\bar{z}) \equiv T_{\bar{z}\bar{z}}$ (depends only on $\bar{z}$)
- **Trace:** $T_{z\bar{z}} = 0$ due to conformal symmetry

Why is $T(z)$ so important? In 2D CFT, the holomorphic stress tensor is the generator of conformal transformations. Its operator product expansion (OPE) with primary fields determines their scaling dimensions and how they transform under conformal maps.

4.3.3 Stress Tensor in the Replica Trick Context

In the replica trick, we consider the partition function Z_n on the n-sheeted Riemann surface $\mathcal{R}_n$. The stress tensor expectation value $\langle T(z) \rangle_{\mathcal{R}_n}$ tells us how the geometry of $\mathcal{R}_n$ affects the energy distribution.

Physical picture: The branch points at u and v act as conical singularities. The field theory responds to this singular geometry by producing a nonzero vacuum expectation value for $T(z)$, even though on the flat plane $\langle T(z) \rangle_{\mathbb{C}} = 0$.

This expectation value is *universal*—it depends only on the central charge c and the positions of the branch points, not on the details of the CFT.

4.3.4 Transformation Law and the Schwarzian

Under a holomorphic change of coordinates $z \mapsto w(z)$, the stress tensor does not transform as a simple tensor (i.e., not just by $(dw/dz)^2$). Instead, it picks up an *anomalous term* proportional to the central charge c:

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\} \tag{187}$$

where $\{w, z\}$ is the Schwarzian derivative.

Interpretation:

- **First term:** The usual tensor transformation (for a primary field of dimension 2)
- **Second term:** The conformal anomaly, which arises because the vacuum is not invariant under general conformal transformations when $c \neq 0$

The Schwarzian derivative $\{w, z\}$ measures the deviation of $w(z)$ from a Möbius transformation (for which the anomaly vanishes).

In the replica trick: We choose $w(z)$ to be the uniformizing map that flattens the n-sheeted surface to a plane. On that plane, the theory is in the standard vacuum with $\langle T(w)\rangle = 0$. Therefore, the expectation value on $\mathcal{R}_n$ comes entirely from the Schwarzian term:

$$\langle T(z)\rangle_{\mathcal{R}_n} = \frac{c}{12}\{w, z\} \tag{188}$$

4.3.5 Why Do We Care About $\langle T(z)\rangle$?

- **Derivative of the free energy:** The stress tensor expectation value is the functional derivative of $\log Z_n$ with respect to the metric. Knowing $\langle T(z)\rangle$ allows us to compute how Z_n depends on the positions u, v.
- **Twist field dimensions:** The singularity of $\langle T(z)\rangle$ near the branch points matches the Ward identity for twist field insertions, giving their scaling dimensions $\Delta_n = \frac{c}{24}(n - 1/n)$.
- **Entanglement entropy:** Ultimately, the derivative $\partial_n \log Z_n|_{n=1}$ gives the entanglement entropy. The stress tensor expectation value provides an intermediate step in this calculation.

4.4 Primary Fields in Conformal Field Theory

We build the concept of a primary field in conformal field theory (CFT) completely from the ground up. We assume knowledge of basic quantum field theory (operators, correlation functions, path integrals) but nothing about CFT specifically.

Quantity	Physical Meaning
$T(z)$	Holomorphic component of energy-momentum tensor
$\langle T(z)\rangle_{\mathcal{R}_n}$	Response to conical singularities at u, v
$\{w, z\}$	Measure of deviation from Möbius transformation
c	Central charge (counts degrees of freedom)

Table 5: Key quantities in the stress tensor formalism.

4.4.1 Why Do We Care About Primary Fields?

In any quantum field theory, we study local operators $\mathcal{O}(x)$ that create or annihilate particles at a spacetime point. Their correlation functions $\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n)\rangle$ encode all physical observables.

In a conformal field theory, the theory is invariant under conformal transformations – angle-preserving maps of spacetime. In two dimensions (one space + one time, Euclidean signature), conformal symmetry is especially powerful because the group of conformal transformations is infinite-dimensional (the Virasoro algebra). This huge symmetry allows us to solve many CFTs exactly.

Primary fields are the simplest, most fundamental local operators in a CFT. They are the building blocks from which all other operators (descendants) are constructed. Understanding primary fields is essential because:

- Their correlation functions are severely constrained by conformal symmetry.
- They determine the entire operator content of the theory.
- They control how the theory responds to changes in geometry (like the replica trick for entanglement entropy).

4.4.2 The 2D Conformal Zoo: Coordinates and Transformations

We work in two-dimensional Euclidean spacetime, with coordinates (x, τ). It is incredibly useful to combine them into complex coordinates:

$$z = x + i\tau, \qquad \bar{z} = x - i\tau. \tag{189}$$

A conformal transformation is a change of coordinates $z \mapsto w(z)$ that preserves angles. In 2D, any holomorphic (complex analytic) function $w(z)$ generates a local conformal transformation. (We also have anti-holomorphic transformations $\bar{w}(\bar{z})$, but we'll focus on the holomorphic sector for now.)

Under $z \mapsto w(z)$, the metric changes by a Weyl factor: $ds^2 = dz d\bar{z} \to \left|\frac{dw}{dz}\right|^2 dw d\bar{w}$. The theory is invariant if the fields transform appropriately.

4.4.3 Primary Fields: The Transformation Law

A primary field $\Phi(z, \bar{z})$ of conformal weights $(h, \bar{h})$ is a local operator that transforms under $z \mapsto w(z)$ as:

$$\boxed{\Phi(z, \bar{z}) = \left(\frac{dw}{dz}\right)^h \left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{h}} \Phi'(w, \bar{w})}. \tag{4}$$

Here:

- h is the holomorphic scaling dimension.
- $\bar{h}$ is the anti-holomorphic scaling dimension.
- The total scaling dimension is $\Delta = h + \bar{h}$, and the spin is $s = h - \bar{h}$.

This is the finite transformation law. It says: if you know the field in the w-coordinates, you get the field in the z-coordinates simply by multiplying by the Jacobian factors. This is the direct generalization of how a tensor component transforms.

Example: A scalar field has $h = \bar{h} = 0$, so it is invariant: $\Phi(z, \bar{z}) = \Phi'(w, \bar{w})$.

Important: The transformation (4) only holds for primary fields. Other fields (descendants) have extra terms involving derivatives of the map.

4.4.4 Infinitesimal Version and the Stress Tensor OPE

An equivalent, often more useful, definition comes from the operator product expansion (OPE) with the stress-energy tensor. The holomorphic stress tensor $T(z)$ is the conserved current for conformal symmetry. Its OPE with a primary field $\Phi(w, \bar{w})$ of weight h is:

$$\boxed{T(z)\Phi(w, \bar{w}) \sim \frac{h}{(z-w)^2}\Phi(w, \bar{w}) + \frac{1}{z-w}\partial_w \Phi(w, \bar{w}) + \text{regular.}} \tag{5}$$

The OPE tells us the singular behavior when two operators get close together. For a primary field, the OPE with $T(z)$ has only a double pole (coefficient h) and a simple pole (coefficient 1, times the derivative). No higher singularities like $1/(z-w)^3$ appear. If they did, the field would not be primary.

Why does this imply the transformation law? The stress tensor $T(z)$ generates conformal transformations. The commutator of a conformal charge $Q_\epsilon = \oint \epsilon(z)T(z)$ with Φ is given by the OPE. Using the OPE (5), one can derive the infinitesimal version of (4), and then integrate to get the finite transformation.

4.4.5 Algebraic Definition: Highest Weight States

There is a powerful algebraic formulation using Virasoro modes. Expand the stress tensor in a Laurent series:

$$T(z) = \sum_{n=-\infty}^{\infty} L_n z^{-n-2}, \qquad L_n = \oint \frac{dz}{2\pi i} z^{n+1} T(z). \tag{190}$$

The modes L_n satisfy the Virasoro algebra:

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}. \tag{191}$$

In this language, the vacuum $|0\rangle$ is annihilated by L_n for $n \geq -1$. A primary state $|\Phi\rangle = \Phi(0,0)|0\rangle$ is defined by:

$$L_0|\Phi\rangle = h|\Phi\rangle, \qquad L_n|\Phi\rangle = 0 \quad \text{for all } n > 0. \tag{192}$$

This means the state has a definite scaling dimension h and is annihilated by all "lowering" operators $L_{n>0}$. It is the highest weight state of a Virasoro representation.

Descendant states are obtained by acting with L_{-n} $(n > 0)$ on the primary state, e.g., $L_{-1}|\Phi\rangle$ corresponds to the derivative $\partial\Phi$.

4.4.6 Quasi-Primary Fields: The Stress Tensor Itself

The stress tensor $T(z)$ is not a primary field. Its OPE with itself is:

$$T(z)T(0) \sim \frac{c/2}{z^4} + \frac{2T(0)}{z^2} + \frac{\partial T(0)}{z} + \text{regular}. \tag{193}$$

The $1/z^4$ term is the central charge anomaly. Under a finite conformal map, $T(z)$ transforms as:

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\}, \tag{194}$$

where $\{w, z\}$ is the Schwarzian derivative. The extra inhomogeneous term means $T(z)$ is *quasi-primary* – it behaves like a primary under global Möbius transformations (for which $\{w, z\} = 0$), but not under general local conformal maps.

4.4.7 Conformal Ward Identity and Correlation Functions

The true power of primary fields is that their correlation functions are completely determined (up to constants) by conformal symmetry.

- **2-point function:** For two primaries with the same dimensions $(h, \bar{h})$,

$$\langle \Phi_1(z_1, \bar{z}_1)\Phi_2(z_2, \bar{z}_2)\rangle = \frac{C_{12}}{|z_1 - z_2|^{2(h+\bar{h})}}. \tag{195}$$

  The functional form is fixed; only the constant C_{12} depends on normalization.

- **3-point function:** The coordinate dependence is fixed up to a constant C_{123} (the OPE coefficient):

$$\langle \Phi_1(z_1)\Phi_2(z_2)\Phi_3(z_3)\rangle = \frac{C_{123}}{|z_{12}|^{\Delta_1+\Delta_2-\Delta_3}|z_{23}|^{\Delta_2+\Delta_3-\Delta_1}|z_{31}|^{\Delta_3+\Delta_1-\Delta_2}}. \tag{196}$$

- **Higher-point functions** are determined by conformal blocks, which are universal functions of the cross-ratios.

This is why CFT is so powerful: the entire theory is encoded in a discrete set of primary fields and their OPE coefficients.

4.4.8 Examples of Primary Fields

Free Massless Boson: Action $S = \frac{1}{2\pi}\int \partial\phi\bar{\partial}\phi$.

- The field ϕ itself is not primary (its OPE with T gives a logarithmic term).
- Its derivative $\partial\phi$ is primary with $h = 1, \bar{h} = 0$.
- Vertex operators $V_\alpha =: e^{i\alpha\phi} :$ are primary with $h = \bar{h} = \alpha^2/2$.

Free Massless Fermion: The holomorphic fermion $\psi(z)$ has OPE $T(z)\psi(0) \sim \frac{1/2}{z^2}\psi(0) + \frac{1}{z}\partial\psi(0)$, so it is primary with $h = 1/2, \bar{h} = 0$.

Minimal Models (e.g., Ising Model): The Ising model has three primary fields:

- The identity $\mathbb{I}$ ($h = 0$)
- The spin field σ ($h = 1/16$)
- The energy density ϵ ($h = 1/2$)

4.4.9 Primary Fields in the Replica Trick

In the replica trick, we introduced twist fields $\mathcal{T}_n$ and $\tilde{\mathcal{T}}_n$. These are primary fields in the orbifold CFT $\mathrm{CFT}^{\otimes n}/\mathbb{Z}_n$. Their scaling dimension is

$$h_n = \bar{h}_n = \frac{c}{24}\left(n - \frac{1}{n}\right). \tag{197}$$

Because they are primary, their two-point function on the plane is fixed:

$$\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = \frac{c_n}{|u-v|^{4h_n}}. \tag{198}$$

This is exactly the partition function ratio $Z_n/(Z_1)^n$. The simple transformation of primary fields under the uniformizing map is what allows us to compute the entanglement entropy.

4.4.10 Summary: What Makes a Field Primary?

- **Transformation:** It picks up only Jacobian factors under conformal maps.
- **OPE with T:** Only a double pole and a simple pole; no higher singularities.
- **Algebraic:** It corresponds to a highest weight state of the Virasoro algebra.
- **Correlators:** Their coordinate dependence is strongly constrained by conformal symmetry.

Primary fields are the "elementary particles" of CFT. Every other local operator can be built by acting on them with derivatives (descendants). Understanding them is the key to unlocking the power of 2D conformal field theory.

4.5 Derivation of the Anomalous Transformation Law

We derive the anomalous transformation law of the holomorphic stress tensor $T(z)$ in 2D conformal field theory from first principles. This includes:

1. Defining the stress tensor and its OPE
2. Deriving the infinitesimal conformal transformation of $T(z)$ from the OPE
3. Exponentiating to finite transformations and showing the appearance of the Schwarzian derivative
4. Proving that the Schwarzian satisfies the required cocycle condition

4.5.1 Stress Tensor and Its OPE

In a 2D Euclidean CFT, the holomorphic stress tensor $T(z)$ is a local field of scaling dimension 2. Its mode expansion on the plane is:

$$T(z) = \sum_{n=-\infty}^{\infty} L_n z^{-n-2}, \tag{199}$$

with the modes satisfying the **Virasoro algebra**:

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12} m(m^2-1)\delta_{m+n,0}. \tag{1}$$

The central charge c appears as an anomalous term. The OPE of $T(z)$ with itself encodes this algebra:

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \text{regular}. \tag{2}$$

The stress tensor is the generator of conformal transformations. For an infinitesimal holomorphic transformation $z \to z + \epsilon v(z)$, the corresponding charge is:

$$Q_\epsilon = \frac{1}{2\pi i} \oint dz\, \epsilon(z) T(z), \tag{200}$$

where the contour encircles the origin (or the points of interest). The variation of any local field $\Phi(w, \bar{w})$ is given by the commutator:

$$\delta_\epsilon \Phi(w) = -[Q_\epsilon, \Phi(w)]. \tag{201}$$

This commutator is evaluated by deforming the contour integral of $T(z)$ around $\Phi(w)$, which picks up the singular terms in the OPE of T with Φ.

4.5.2 Infinitesimal Transformation of $T(z)$ from OPE

We apply this to the stress tensor itself: $\Phi(w) = T(w)$. The OPE (2) gives:

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w}. \tag{202}$$

The commutator is:

$$[Q_\epsilon, T(w)] = \frac{1}{2\pi i}\oint_w dz\, \epsilon(z)T(z)T(w), \tag{203}$$

where the contour circles w counterclockwise. Using the OPE, we get contributions from each singular term.

Systematic computation: Expand $\epsilon(z)$ around w:

$$\epsilon(z) = \epsilon(w) + \epsilon'(w)(z-w) + \frac{1}{2}\epsilon''(w)(z-w)^2 + \frac{1}{6}\epsilon'''(w)(z-w)^3 + \cdots. \tag{204}$$

Term 1: $\frac{c/2}{(z-w)^4}$

$$\frac{\epsilon(z)}{(z-w)^4} = \frac{\epsilon(w)}{(z-w)^4} + \frac{\epsilon'(w)}{(z-w)^3} + \frac{\epsilon''(w)}{2(z-w)^2} + \frac{\epsilon'''(w)}{6(z-w)} + \cdots. \tag{205}$$

The residue is $\frac{1}{6}\epsilon'''(w)$. The contribution is:

$$\frac{c/2}{2\pi i}\oint dz \frac{\epsilon(z)}{(z-w)^4} = \frac{c}{12}\partial^3\epsilon(w). \tag{206}$$

Term 2: $\frac{2T(w)}{(z-w)^2}$

$$\frac{\epsilon(z)}{(z-w)^2} = \frac{\epsilon(w)}{(z-w)^2} + \frac{\epsilon'(w)}{z-w} + \cdots. \tag{207}$$

The residue is $\epsilon'(w)$. The contribution is:

$$2T(w)\epsilon'(w). \tag{208}$$

Term 3: $\frac{\partial_w T(w)}{z-w}$

$$\frac{\epsilon(z)}{z-w} = \frac{\epsilon(w)}{z-w} + \epsilon'(w) + \cdots. \tag{209}$$

The residue is $\epsilon(w)$. The contribution is:

$$\partial_w T(w)\epsilon(w). \tag{210}$$

Result: Putting together:

$$[Q_\epsilon, T(w)] = \frac{c}{12}\partial^3\epsilon(w) + 2\epsilon'(w)T(w) + \epsilon(w)\partial T(w). \tag{211}$$

Since $\delta_\epsilon T = -[Q_\epsilon, T]$, we get:

$$\delta_\epsilon T(w) = -\frac{c}{12}\partial^3\epsilon(w) - 2\epsilon'(w)T(w) - \epsilon(w)\partial T(w). \tag{3}$$

With the standard sign convention (active transformation $z \to z + \epsilon(z)$), this is usually written:

$$\delta_\epsilon T(z) = \frac{c}{12}\partial^3\epsilon(z) + 2\partial\epsilon(z)T(z) + \epsilon(z)\partial T(z). \tag{212}$$

Equation (3) is the **infinitesimal transformation of** $T(z)$.

4.5.3 From Infinitesimal to Finite Transformation

We now derive the finite transformation under a holomorphic map $z \mapsto w(z)$. Suppose $w(z)$ is a finite conformal transformation, obtained by applying a sequence of infinitesimal transformations.

Ansatz: The finite transformation law has the form:

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12} S(w, z), \tag{213}$$

where $S(w, z)$ is some function to be determined. We will show that $S(w, z) = \{w, z\}$, the Schwarzian derivative.

Check: Expand to first order. Let $w(z) = z + \epsilon v(z)$. Then:

- $\frac{dw}{dz} = 1 + \epsilon v'(z)$
- $\left(\frac{dw}{dz}\right)^2 = 1 + 2\epsilon v'(z) + O(\epsilon^2)$
- $T(w) = T(z + \epsilon v) = T(z) + \epsilon v(z)\partial T(z) + O(\epsilon^2)$

Compute the Schwarzian for $w = z + \epsilon v$:

$$w' = 1 + \epsilon v', \quad w'' = \epsilon v'', \quad w''' = \epsilon v''', \tag{214}$$

$$\frac{w''}{w'} = \epsilon v''(1 - \epsilon v' + \cdots) \approx \epsilon v'', \tag{215}$$

$$\left(\frac{w''}{w'}\right)^2 \approx \epsilon^2 (v'')^2 \text{ (higher order)}, \tag{216}$$

$$\frac{w'''}{w'} \approx \epsilon v'''. \tag{217}$$

Thus $\{w, z\} = \epsilon v''' + O(\epsilon^2)$.

Plugging into the transformation:

$$T(z) = (1 + 2\epsilon v')\,(T(z) + \epsilon v \partial T) + \frac{c}{12}\epsilon v''' + O(\epsilon^2) \tag{218}$$

$$= T(z) + \epsilon v \partial T(z) + 2\epsilon v' T(z) + \frac{c}{12}\epsilon v''' + O(\epsilon^2). \tag{219}$$

Cancel $T(z)$ from both sides, divide by ϵ:

$$0 = v\partial T + 2v'T + \frac{c}{12} v'''. \tag{220}$$

Rearranging gives the active change:

$$\delta T = -\epsilon \left(v\partial T + 2v'T + \frac{c}{12} v'''\right), \tag{221}$$

which matches Eq. (3) with $\epsilon(z) = \epsilon v(z)$. **The finite transformation with the Schwarzian derivative correctly reproduces the infinitesimal OPE result.**

4.5.4 The Cocycle Condition and Uniqueness

The above shows that if a finite transformation exists, it must contain the Schwarzian term. We now prove that the transformation law is consistent under composition of maps, which forces the anomalous term to be the Schwarzian derivative.

Requirement: If we perform two successive transformations $z \to w \to \xi$, the result should be the same as a single transformation $z \to \xi$.

Direct transformation $z \to \xi$:

$$T(z) = \left(\frac{d\xi}{dz}\right)^2 T(\xi) + \frac{c}{12}\{\xi, z\}. \tag{222}$$

Via intermediate w:

$$T(w) = \left(\frac{d\xi}{dw}\right)^2 T(\xi) + \frac{c}{12}\{\xi, w\}, \tag{223}$$

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\}. \tag{224}$$

Substitute $T(w)$:

$$T(z) = \left(\frac{dw}{dz}\right)^2 \left[\left(\frac{d\xi}{dw}\right)^2 T(\xi) + \frac{c}{12}\{\xi, w\}\right] + \frac{c}{12}\{w, z\} \tag{225}$$

$$= \left(\frac{d\xi}{dz}\right)^2 T(\xi) + \frac{c}{12}\left(\{\xi, w\}\left(\frac{dw}{dz}\right)^2 + \{w, z\}\right). \tag{226}$$

where we used the chain rule $\frac{d\xi}{dz} = \frac{d\xi}{dw}\frac{dw}{dz}$.

Consistency requires:

$$\{\xi, z\} = \{\xi, w\}\left(\frac{dw}{dz}\right)^2 + \{w, z\}. \tag{4}$$

This is precisely the **chain rule for the Schwarzian derivative**! Thus, the Schwarzian is the unique function (up to normalization) that satisfies this cocycle condition and reduces to the third derivative for infinitesimal maps.

Conclusion: The anomalous transformation law is:

$$\boxed{T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\}} \tag{227}$$

where $\{w, z\} = \frac{w'''}{w'} - \frac{3}{2}\left(\frac{w''}{w'}\right)^2$.

4.5.5 Alternative Derivation via Virasoro Generators

Another rigorous method is to examine how the Virasoro modes transform under a conformal map. On the plane, the stress tensor is $T(z) = \sum L_n z^{-n-2}$. Under a map $z \mapsto w(z)$, we can

define a new stress tensor $T'(w)$ such that the physics remains the same. The transformation of the modes can be computed and leads to the same Schwarzian derivative term. The standard result is:

$$T(z) = \left(\frac{dw}{dz}\right)^2 T'(w) + \frac{c}{12}\{w, z\}. \tag{228}$$

Step	**Result**
Stress tensor OPE	$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}$
Infinitesimal transformation	$\delta_\epsilon T = \frac{c}{12}\partial^3\epsilon + 2\partial\epsilon\, T + \epsilon\partial T$
Finite transformation	$T(z) = (dw/dz)^2 T(w) + \frac{c}{12}\{w, z\}$
Cocycle condition	$\{\xi, z\} = \{\xi, w\}(dw/dz)^2 + \{w, z\}$

Table 6: Summary of the derivation of the anomalous transformation law.

4.5.6 The Virasoro Generators L_n

In the context of 2D conformal field theory, L_n (and similarly $\bar{L}_n$) refers to the **Virasoro generators**—the Fourier modes of the holomorphic stress-energy tensor $T(z)$.

Mode expansion of the stress tensor On the complex plane (or cylinder), the holomorphic stress tensor $T(z)$ is a local field of scaling dimension 2. We can expand it in a Laurent series:

$$T(z) = \sum_{n=-\infty}^{\infty} L_n\, z^{-n-2}. \tag{1}$$

The coefficients L_n are operators acting on the Hilbert space of the CFT. The inverse relation is:

$$L_n = \frac{1}{2\pi i}\oint dz\, z^{n+1} T(z), \tag{2}$$

where the contour encircles the origin.

Physical interpretation: L_n generates the n-th mode of conformal transformations. Specifically:

- $L_0 + \bar{L}_0$ is proportional to the Hamiltonian (energy operator)
- L_{-1} and $\bar{L}_{-1}$ generate translations
- L_1, L_0, L_{-1} form an $\mathfrak{sl}(2, \mathbb{C})$ subalgebra that generates global conformal transformations (Möbius maps)
- Higher $|n|$ modes generate local, non-global conformal transformations

The Virasoro algebra The modes satisfy the **Virasoro algebra**:

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}. \tag{3}$$

- The first term $(m-n)L_{m+n}$ is what one would get from the classical Poisson brackets of the stress tensor (the Witt algebra)
- The second term, proportional to the central charge c, is the *quantum anomaly.* It arises because the stress tensor OPE contains the $c/2$ term, and it is ultimately responsible for the Schwarzian derivative in the finite transformation law

Role in the transformation law When we performed an infinitesimal conformal transformation $z \to z + \epsilon v(z)$ and expanded $\epsilon(z) = \sum \epsilon_n z^{n+1}$, the corresponding charge is:

$$Q_\epsilon = \frac{1}{2\pi i}\oint dz\, \epsilon(z)T(z) = \sum_n \epsilon_n L_n. \tag{229}$$

The commutator $[Q_\epsilon, T(w)]$ generates the variation of the stress tensor. This directly leads to the infinitesimal transformation:

$$\delta_\epsilon T(w) = \frac{c}{12}\partial^3\epsilon(w) + 2\partial\epsilon(w)T(w) + \epsilon(w)\partial T(w), \tag{230}$$

which integrates to the finite Schwarzian transformation.

Connection to the replica trick In the replica trick, we consider the CFT on the n-sheeted Riemann surface $\mathcal{R}_n$. The uniformizing map $w(z)$ is not a global conformal transformation (it is not Möbius), so the modes L_n on the z-plane and the w-plane are related by a nontrivial Bogoliubov transformation. The anomalous term $\frac{c}{12}\{w, z\}$ is exactly the finite version of the central term in the Virasoro algebra.

In practice, when computing $\langle T(z)\rangle_{\mathcal{R}_n}$, we don't need to explicitly use the modes L_n; we work directly with the transformation law. But the modes underpin the entire structure.

Symbol	**Meaning**
$T(z)$	Holomorphic stress tensor (local field)
L_n	Mode operators from Laurent expansion of $T(z)$
c	Central charge of the CFT
$[L_m, L_n]$	Virasoro algebra, including central extension

Table 7: Virasoro generators and their role in 2D CFT.

Summary: The L_n are the fundamental symmetry generators in 2D CFT. Their algebra, especially the central charge term, is the algebraic origin of the Schwarzian derivative that appears in the replica trick computation of entanglement entropy.

4.6 Primary Fields in Conformal Field Theory

We build the concept of a primary field in conformal field theory (CFT) from the ground up. We assume familiarity with basic quantum field theory (operators, correlation functions, path integrals) but nothing about CFT specifically.

4.6.1 Why Do We Care About Primary Fields?

In any quantum field theory, we study local operators $\mathcal{O}(x)$ that create or annihilate particles at a spacetime point. Their correlation functions $\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle$ encode all physical observables.

In a conformal field theory, the theory is invariant under conformal transformations – angle-preserving maps of spacetime. In two dimensions (one space + one time, Euclidean signature), conformal symmetry is especially powerful because the group of conformal transformations is infinite-dimensional (the Virasoro algebra). This huge symmetry allows us to solve many CFTs exactly.

Primary fields are the simplest, most fundamental local operators in a CFT. They are the building blocks from which all other operators (descendants) are constructed. Understanding primary fields is essential because:

- Their correlation functions are severely constrained by conformal symmetry
- They determine the entire operator content of the theory
- They control how the theory responds to changes in geometry (like the replica trick for entanglement entropy)

4.6.2 The 2D Conformal Zoo: Coordinates and Transformations

We work in two-dimensional Euclidean spacetime, with coordinates (x, τ). It is incredibly useful to combine them into complex coordinates:

$$z = x + i\tau, \qquad \bar{z} = x - i\tau. \tag{231}$$

A conformal transformation is a change of coordinates $z \mapsto w(z)$ that preserves angles. In 2D, any holomorphic (complex analytic) function $w(z)$ generates a local conformal transformation. (We also have anti-holomorphic transformations $\bar{w}(\bar{z})$, but we focus on the holomorphic sector for now.)

Under $z \mapsto w(z)$, the metric changes by a Weyl factor: $ds^2 = dz d\bar{z} \to \left|\frac{dw}{dz}\right|^2 dw d\bar{w}$. The theory is invariant if the fields transform appropriately.

4.6.3 Primary Fields: The Transformation Law

A primary field $\Phi(z, \bar{z})$ of conformal weights $(h, \bar{h})$ is a local operator that transforms under $z \mapsto w(z)$ as:

$$\boxed{\Phi(z, \bar{z}) = \left(\frac{dw}{dz}\right)^h \left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{h}} \Phi'(w, \bar{w})}. \tag{1}$$

Here:

- h is the holomorphic scaling dimension
- $\bar{h}$ is the anti-holomorphic scaling dimension
- The total scaling dimension is $\Delta = h + \bar{h}$, and the spin is $s = h - \bar{h}$

This is the finite transformation law. It says: if you know the field in the w-coordinates, you get the field in the z-coordinates simply by multiplying by the Jacobian factors. This is the direct generalization of how a tensor component transforms.

Example: A scalar field has $h = \bar{h} = 0$, so it is invariant: $\Phi(z, \bar{z}) = \Phi'(w, \bar{w})$.

Important: The transformation (1) only holds for primary fields. Other fields (descendants) have extra terms involving derivatives of the map.

4.6.4 Infinitesimal Version and the Stress Tensor OPE

An equivalent, often more useful, definition comes from the operator product expansion (OPE) with the stress-energy tensor. The holomorphic stress tensor $T(z)$ is the conserved current for conformal symmetry. Its OPE with a primary field $\Phi(w, \bar{w})$ of weight h is:

$$\boxed{T(z)\Phi(w, \bar{w}) \sim \frac{h}{(z-w)^2}\Phi(w, \bar{w}) + \frac{1}{z-w}\partial_w \Phi(w, \bar{w}) + \text{regular.}} \tag{2}$$

The OPE tells us the singular behavior when two operators get close together. For a primary field, the OPE with $T(z)$ has only a double pole (coefficient h) and a simple pole (coefficient 1, times the derivative). No higher singularities like $1/(z-w)^3$ appear. If they did, the field would not be primary.

Why does this imply the transformation law? The stress tensor $T(z)$ generates conformal transformations. The commutator of a conformal charge $Q_\epsilon = \oint \epsilon(z)T(z)$ with Φ is given by the OPE. Using the OPE (2), one can derive the infinitesimal version of (1), and then integrate to get the finite transformation.

4.6.5 Algebraic Definition: Highest Weight States

There is a powerful algebraic formulation using Virasoro modes. Expand the stress tensor in a Laurent series:

$$T(z) = \sum_{n=-\infty}^{\infty} L_n z^{-n-2}, \qquad L_n = \oint \frac{dz}{2\pi i} z^{n+1} T(z). \tag{232}$$

The modes L_n satisfy the Virasoro algebra:

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}. \tag{233}$$

In this language, the vacuum $|0\rangle$ is annihilated by L_n for $n \geq -1$. A primary state $|\Phi\rangle = \Phi(0,0)|0\rangle$ is defined by:

$$L_0|\Phi\rangle = h|\Phi\rangle, \qquad L_n|\Phi\rangle = 0 \quad \text{for all } n > 0. \tag{234}$$

This means the state has a definite scaling dimension h and is annihilated by all "lowering" operators $L_{n>0}$. It is the highest weight state of a Virasoro representation.

Descendant states are obtained by acting with L_{-n} $(n > 0)$ on the primary state, e.g., $L_{-1}|\Phi\rangle$ corresponds to the derivative $\partial\Phi$.

4.6.6 Quasi-Primary Fields: The Stress Tensor Itself

The stress tensor $T(z)$ is not a primary field. Its OPE with itself is:

$$T(z)T(0) \sim \frac{c/2}{z^4} + \frac{2T(0)}{z^2} + \frac{\partial T(0)}{z} + \text{regular}. \tag{235}$$

The $1/z^4$ term is the central charge anomaly. Under a finite conformal map, $T(z)$ transforms as:

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\}, \tag{236}$$

where $\{w, z\}$ is the Schwarzian derivative. The extra inhomogeneous term means $T(z)$ is **quasi-primary** – it behaves like a primary under global Möbius transformations (for which $\{w, z\} = 0$), but not under general local conformal maps.

4.6.7 Conformal Ward Identity and Correlation Functions

The true power of primary fields is that their correlation functions are completely determined (up to constants) by conformal symmetry.

2-point function: For two primaries with the same dimensions $(h, \bar{h})$,

$$\langle \Phi_1(z_1, \bar{z}_1)\Phi_2(z_2, \bar{z}_2)\rangle = \frac{C_{12}}{|z_1 - z_2|^{2(h+\bar{h})}}. \tag{237}$$

The functional form is fixed; only the constant C_{12} depends on normalization.

3-point function: The coordinate dependence is fixed up to a constant C_{123} (the OPE coefficient):

$$\langle \Phi_1(z_1)\Phi_2(z_2)\Phi_3(z_3)\rangle = \frac{C_{123}}{|z_{12}|^{\Delta_1+\Delta_2-\Delta_3}|z_{23}|^{\Delta_2+\Delta_3-\Delta_1}|z_{31}|^{\Delta_3+\Delta_1-\Delta_2}}. \tag{238}$$

Higher-point functions are determined by conformal blocks, which are universal functions of the cross-ratios.

This is why CFT is so powerful: the entire theory is encoded in a discrete set of primary fields and their OPE coefficients.

4.6.8 Examples of Primary Fields

Free Massless Boson Action: $S = \frac{1}{2\pi}\int \partial\phi\bar{\partial}\phi$. The field ϕ itself is not primary (its OPE with T gives a logarithmic term). But its derivative $\partial\phi$ is primary with $h = 1, \bar{h} = 0$. Vertex operators $V_\alpha =: e^{i\alpha\phi} :$ are primary with $h = \bar{h} = \alpha^2/2$.

Free Massless Fermion The holomorphic fermion $\psi(z)$ has OPE $T(z)\psi(0) \sim \frac{1/2}{z^2}\psi(0) + \frac{1}{z}\partial\psi(0)$, so it is primary with $h = 1/2, \bar{h} = 0$.

Minimal Models (e.g., Ising Model) The Ising model has three primary fields: the identity $\mathbb{I}$ ($h = 0$), the spin field σ ($h = 1/16$), and the energy density ϵ ($h = 1/2$).

4.6.9 Primary Fields in the Replica Trick

In the replica trick, we introduced twist fields $\mathcal{T}_n$ and $\tilde{\mathcal{T}}_n$. These are primary fields in the orbifold CFT $\mathrm{CFT}^{\otimes n}/\mathbb{Z}_n$. Their scaling dimension is

$$h_n = \bar{h}_n = \frac{c}{24}\left(n - \frac{1}{n}\right). \tag{239}$$

Because they are primary, their two-point function on the plane is fixed:

$$\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = \frac{c_n}{|u-v|^{4h_n}}. \tag{240}$$

This is exactly the partition function ratio $Z_n/(Z_1)^n$. The simple transformation of primary fields under the uniformizing map is what allows us to compute the entanglement entropy.

4.6.10 Summary: What Makes a Field Primary?

- **Transformation:** It picks up only Jacobian factors under conformal maps
- **OPE with T:** Only a double pole and a simple pole; no higher singularities
- **Algebraic:** It corresponds to a highest weight state of the Virasoro algebra
- **Correlators:** Their coordinate dependence is strongly constrained by conformal symmetry

Primary fields are the "elementary particles" of CFT. Every other local operator can be built by acting on them with derivatives (descendants). Understanding them is the key to unlocking the power of 2D conformal field theory.

Property	**Primary Field Behavior**
Finite transformation	$\Phi(z) = (dw/dz)^h (d\bar{w}/d\bar{z})^{\bar{h}}\Phi'(w)$
OPE with $T(z)$	$\frac{h}{(z-w)^2}\Phi(w) + \frac{1}{z-w}\partial\Phi(w)$
Virasoro action	$L_0\lvert\Phi\rangle = h\lvert\Phi\rangle$, $L_{n>0}\lvert\Phi\rangle = 0$
2-point function	$\langle\Phi_1\Phi_2\rangle \propto \lvert z_{12}\rvert^{-2\Delta}$

Table 8: Summary of primary field properties in 2D CFT.

4.7 Stress Tensor Transformation

Under a conformal transformation $z \to w(z)$, the stress tensor transforms as:

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\} \tag{241}$$

where $\{w, z\}$ is the Schwarzian derivative:

$$\{w, z\} = \frac{w'''}{w'} - \frac{3}{2}\left(\frac{w''}{w'}\right)^2 \tag{242}$$

Compute derivatives of $w(z)$:

$$w' = \frac{dw}{dz} = \frac{1}{n}\left(\frac{z-u}{z-v}\right)^{\frac{1}{n}-1} \frac{v-u}{(z-v)^2} \tag{243}$$

$$w'' = \frac{d^2w}{dz^2} = \frac{1}{n}\left(\frac{1}{n}-1\right)\left(\frac{z-u}{z-v}\right)^{\frac{1}{n}-2} \frac{(v-u)^2}{(z-v)^4} - \frac{2}{n}\left(\frac{z-u}{z-v}\right)^{\frac{1}{n}-1} \frac{v-u}{(z-v)^3} \tag{244}$$

Schwarzian derivative (after simplification):

$$\{w, z\} = \frac{n^2-1}{2n^2} \frac{(v-u)^2}{(z-u)^2(z-v)^2} \tag{245}$$

4.8 Scaling Dimension of Twist Fields

The expectation value of the stress tensor in the presence of twist fields is:

$$\langle T(z)\rangle_{\mathcal{R}_n} = \frac{c}{12}\{w, z\} \tag{246}$$

Compare to CFT Ward identity: For primary fields $\mathcal{T}_n(u)$ and $\tilde{\mathcal{T}}_n(v)$:

$$\langle T(z)\mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = \left[\frac{h_n}{(z-u)^2} + \frac{h_n}{(z-v)^2} + \frac{1}{z-u}\partial_u + \frac{1}{z-v}\partial_v\right]\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle \tag{247}$$

Match the singular terms: Comparing Eq. (47) with the Schwarzian (Eq. 46):

$$\frac{c}{12} \cdot \frac{n^2-1}{2n^2} \frac{(v-u)^2}{(z-u)^2(z-v)^2} = \frac{h_n}{(z-u)^2} + \frac{h_n}{(z-v)^2} + \cdots \tag{248}$$

Extract scaling dimension:

$$h_n = \frac{c}{24}\left(n - \frac{1}{n}\right) \tag{249}$$

The full scaling dimension (holomorphic + antiholomorphic) is:

$$\Delta_n = h_n + \bar{h}_n = \frac{c}{12}\left(n - \frac{1}{n}\right) \tag{250}$$

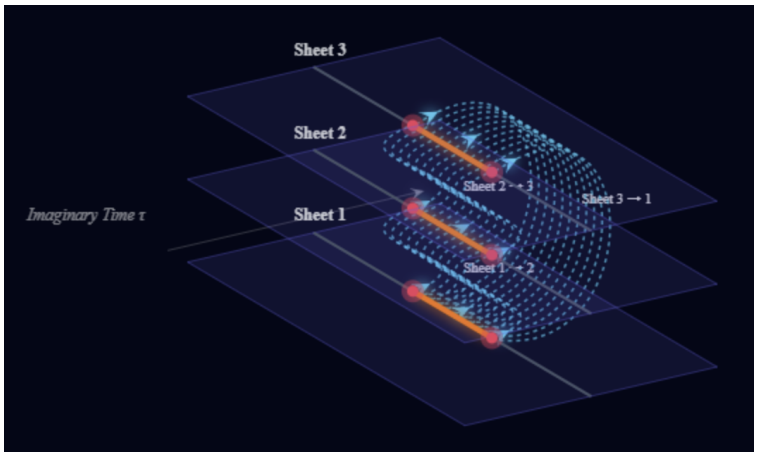


Figure 1: Schematic illustration of the branch cut structure. The cut along interval $A = [u, v]$ creates an n-sheeted Riemann surface with cyclic gluing: sheet j connects to sheet $j+1$ across the cut. The branch points at u and v are where twist fields $\mathcal{T}_n$ and $\tilde{\mathcal{T}}_n$ are inserted.

4.9 Twist Fields as Local Operators in the n-Copy CFT

We explain why and how twist fields are defined as local operators in a multi-copy CFT, and how the partition function on the replica surface is expressed as their correlation function.

4.9.1 The Problem: Non-Local Fields on the Plane

Goal: Compute the partition function $Z_{\mathcal{R}_{n,1}}$ on the n-sheeted Riemann surface (the n-sheeted cover for a single interval $[u, v]$).

First attempt: Map $\mathcal{R}_{n,1}$ to the ordinary complex plane $z = x + i\tau$ using a uniformizing map. However, the Riemann surface has curvature singularities (branch points) at u and v. To mimic this on the flat plane, we must impose special boundary conditions on the fields around those points. In the operator language, these boundary conditions are enforced by inserting some "fields" at u and v.

The catch: If we define these fields directly via the path integral on the plane with those boundary conditions, they turn out to be *non-local* operators. Locality is crucial in QFT (e.g., for OPEs, Ward identities), so this is undesirable.

4.9.2 The Solution: Move Topology from Worldsheet to Target Space

Instead of putting the complicated topology into the *worldsheet* (the space of coordinates x, τ), we keep the worldsheet flat ($\mathbb{C}$) and put the complexity into the *target space*—the space of field values.

The trick: Re-interpret the theory on $\mathcal{R}_{n,1}$ as a theory of n independent copies of the original CFT, but with a specific boundary condition along the cut.

- Let the original fields be $\varphi(x, \tau)$. Introduce n copies: $\varphi_1, \varphi_2, \ldots, \varphi_n$.

- The total Lagrangian is simply the sum of the individual Lagrangians:

$$\mathcal{L}^{(n)}[\varphi_1, \ldots, \varphi_n] = \sum_{j=1}^{n} \mathcal{L}[\varphi_j]. \tag{251}$$

  So the copies are decoupled in the bulk.

- However, we impose the following boundary conditions across the cut $x \in [u, v]$, $\tau = 0$:

$$\varphi_i(x, 0^+) = \varphi_{i+1}(x, 0^-), \qquad i = 1, \ldots, n, \tag{252}$$

  with the identification $\varphi_{n+1} \equiv \varphi_1$.

This precisely reproduces the cyclic gluing of the sheets: going from just below the cut on copy $i+1$ to just above the cut on copy i (or vice versa, depending on the orientation).

The path integral over the n copies on the plane, restricted by these boundary conditions, is exactly the partition function $Z_{\mathcal{R}}$.

4.9.3 Twist Fields as Local Operators Enforcing the Boundary Condition

The restricted path integral with boundary conditions can be represented by inserting local operators at the branch points u and v. These operators are called *twist fields* (or *branch-point twist fields*).

Why are they local in the n-copy model? Because the boundary condition only relates the copies at a specific line segment, and this relation is a *global internal symmetry* of the n-copy model: the cyclic permutation $\sigma : i \mapsto i + 1 \mod n$. The Lagrangian $\mathcal{L}^{(n)}$ is invariant under any permutation of the copy labels.

In any QFT, if you have a global internal symmetry σ, you can define a *twist operator* $\mathcal{T}_\sigma$ that, when inserted, creates a branch cut such that fields undergo the symmetry transformation σ when transported around the insertion point. Here:

- $\mathcal{T}_n \equiv \mathcal{T}_\sigma$ implements the forward cyclic permutation $i \mapsto i + 1$.
- $\tilde{\mathcal{T}}_n \equiv \mathcal{T}_{\sigma^{-1}}$ implements the inverse permutation $i + 1 \mapsto i$.

These are *local operators* in the orbifold theory $\mathcal{L}^{(n)}$, and their insertion at u and v enforces the required boundary condition across the segment $[u, v]$.

4.9.4 The Proportionality Relation

For a general set A consisting of N disjoint intervals $[u_j, v_j]$, the partition function on the n-sheeted Riemann surface is proportional to the correlation function of twist fields:

$$Z_{\mathcal{R}_{n,N}} \propto \langle \mathcal{T}_n(u_1, 0)\tilde{\mathcal{T}}_n(v_1, 0) \cdots \mathcal{T}_n(u_N, 0)\tilde{\mathcal{T}}_n(v_N, 0)\rangle_{\mathcal{L}^{(n)},\mathbb{C}}. \tag{16}$$

- **Left side:** Partition function on the n-sheeted Riemann surface with N cuts.

- **Right side:** Correlation function of twist fields in the n-copy model on the flat complex plane.
- **Proportionality constant:** Independent of the positions u_j, v_j; it only depends on the UV cutoff and the normalization of the twist fields. This constant cancels when we take ratios like $Z_n/(Z_1)^n$ and compute entanglement entropy.

This relation is the foundation for all subsequent CFT calculations of entanglement entropy.

Direct Approach (Problem)	**n-Copy Approach (Solution)**
Map $\mathcal{R}_n \to \mathbb{C}$ directly	Keep worldsheet flat, use n copies
Boundary conditions at u, v are non-local	Boundary condition is global symmetry σ
Operators are non-local on $\mathbb{C}$	Twist fields are local in $\mathcal{L}^{(n)}$
Hard to use OPE/Ward identities	Standard CFT correlator techniques apply

Table 9: Comparison of direct mapping vs. n-copy orbifold approach for defining twist fields.

Summary of the Logical Flow:

1. **Goal:** Compute $Z_{\mathcal{R}}$ on a multi-sheeted surface.
2. **Obstacle:** Direct mapping to the plane makes the required boundary-condition operators non-local.
3. **Resolution:** Use an n-copy model on the plane. The multi-sheeted geometry becomes a flat plane with a permutation boundary condition across the cuts.
4. **Twist fields:** The permutation is a global symmetry of the n-copy model, so we can define local twist fields that create branch cuts implementing this permutation.
5. **Result:** $Z_{\mathcal{R}}$ is proportional to the correlation function of these twist fields on the plane.

This construction is completely general and works for any CFT, interacting or free. It reduces the problem of computing partition functions on complicated surfaces to standard correlators in an orbifold CFT.

4.10 Derivation of the Partition Function Ratio

We derive from first principles why the ratio of partition functions $Z_n/(Z_1)^n$ equals the two-point function of twist fields $\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle$. This is the key bridge between the geometric replica construction and the operator formalism of conformal field theory.

4.10.1 The Partition Functions Z_n and Z_1

- Z_1 is the partition function of the CFT on the ordinary complex plane $\mathbb{C}$ (or a cylinder, etc.). For vacuum entanglement entropy, we take the theory on the plane, so Z_1 is just the vacuum functional integral with no operator insertions.
- Z_n is the partition function on the n-sheeted Riemann surface $\mathcal{R}_n$ obtained by taking n copies of the plane, cutting each along the interval $A = [u, v]$ on the real axis, and cyclically gluing the edges: the lower edge of copy j is glued to the upper edge of copy $j + 1$ (with $n + 1 \equiv 1$).

Both Z_n and Z_1 are formally defined as Euclidean path integrals:

$$Z_n = \int_{\mathcal{R}_n} \mathcal{D}\phi \, e^{-S[\phi]}, \qquad Z_1 = \int_{\mathbb{C}} \mathcal{D}\phi \, e^{-S[\phi]}. \tag{253}$$

4.10.2 Reinterpreting the Multi-Sheeted Geometry as an Orbifold Theory

Instead of working on the curved space $\mathcal{R}_n$, we can map the problem to a theory on the ordinary plane $\mathbb{C}$ by "unwrapping" the sheets. Consider n independent copies of the original CFT, with fields $\phi_1, \phi_2, \ldots, \phi_n$. The total action is the sum of decoupled actions:

$$S_{\text{tot}} = \sum_{j=1}^{n} S[\phi_j]. \tag{254}$$

On the plane, these fields are independent. However, to reproduce the cyclic gluing along the cut $[u, v]$, we must impose boundary conditions on the fields across that interval. Specifically, when crossing the cut from below to above, the fields should transform as:

$$\phi_j(x^+) = \phi_{j+1}(x^-), \quad x \in [u, v], \tag{255}$$

with $\phi_{n+1} \equiv \phi_1$.

These boundary conditions can be enforced by inserting local operators at the endpoints u and v that implement the cyclic permutation of the copies. Such operators are called *twist fields*.

4.10.3 Twist Fields as Boundary Condition Changing Operators

In the operator formulation, a local operator $\mathcal{O}(u)$ is defined by the path integral with specific boundary conditions around the point u. For a standard primary field $\Phi(u)$, the path integral over fields on the plane is modified by a factor of Φ inserted at u. For a twist field $\mathcal{T}_n(u)$, the boundary condition is that as one circles around u, the fields ϕ_j undergo a cyclic permutation: $\phi_j \mapsto \phi_{j+1}$. This is exactly the monodromy that fields experience when going around a branch point on $\mathcal{R}_n$.

Thus, the partition function Z_n on $\mathcal{R}_n$ can be written as a path integral on the plane with two twist field insertions: one at u and one at v. At u, the twist field $\mathcal{T}_n$ creates the

branch point; at v, the anti-twist field $\tilde{\mathcal{T}}_n$ (which implements the inverse permutation) closes the cut. More precisely:

$$Z_n = \int_{\mathbb{C}} \mathcal{D}\phi_1 \cdots \mathcal{D}\phi_n \, e^{-\sum_j S[\phi_j]} \times \text{(boundary conditions from twist fields)}. \tag{256}$$

The boundary conditions are: on a small circle around u, the fields ϕ_j are identified with ϕ_{j+1} after a full rotation. This is exactly the effect of inserting $\mathcal{T}_n(u)$ and $\tilde{\mathcal{T}}_n(v)$.

4.10.4 Normalization and the Ratio $Z_n/(Z_1)^n$

The path integral with twist fields is not normalized. The normalization is provided by the path integral on the plane with no twist fields, which is simply the product of n independent partition functions of the original CFT:

$$\int_{\mathbb{C}} \mathcal{D}\phi_1 \cdots \mathcal{D}\phi_n \, e^{-\sum_j S[\phi_j]} = (Z_1)^n. \tag{257}$$

Therefore, the normalized correlation function of the twist fields is exactly the ratio:

$$\langle \mathcal{T}_n(u) \tilde{\mathcal{T}}_n(v) \rangle_{\mathbb{C}} = \frac{\int_{\mathbb{C}} \mathcal{D}\vec{\phi} \, e^{-S_{\text{tot}}} \text{ (with twist b.c.)}}{\int_{\mathbb{C}} \mathcal{D}\vec{\phi} \, e^{-S_{\text{tot}}}} = \frac{Z_n}{(Z_1)^n}. \tag{1}$$

This is the **fundamental relation**.

4.10.5 Twist Fields as Primary Operators in the Orbifold CFT

The theory of n decoupled copies with a $\mathbb{Z}_n$ cyclic permutation symmetry is an *orbifold CFT*. The twist fields $\mathcal{T}_n$ and $\tilde{\mathcal{T}}_n$ are the primary operators that create the twisted sector. Their conformal dimensions can be determined by the conformal anomaly, as we saw earlier via the uniformizing map and the stress tensor expectation value.

Since they are primary fields with scaling dimension $\Delta_n = \frac{c}{12}(n - 1/n)$, their two-point function on the plane is fixed by conformal invariance up to a constant:

$$\langle \mathcal{T}_n(u) \tilde{\mathcal{T}}_n(v) \rangle = \frac{c_n}{|u-v|^{2\Delta_n}} = c_n \, |u-v|^{-\frac{c}{6}\left(n - \frac{1}{n}\right)}. \tag{2}$$

Equating (1) and (2) gives the **replica trick formula**:

$$\frac{Z_n}{(Z_1)^n} = c_n \, |u-v|^{-\frac{c}{6}\left(n - \frac{1}{n}\right)}. \tag{258}$$

4.10.6 Alternative Derivation via the Uniformizing Map

We can also derive (2) without explicitly constructing the orbifold CFT. The uniformizing map $w(z) = ((z-u)/(z-v))^{1/n}$ maps $\mathcal{R}_n$ to the plane. The stress tensor expectation value on $\mathcal{R}_n$ is given by the Schwarzian:

$$\langle T(z) \rangle_{\mathcal{R}_n} = \frac{c}{12} \{w, z\}. \tag{259}$$

Comparing the singularities of $\langle T(z)\rangle_{\mathcal{R}_n}$ near u and v with the conformal Ward identity for two primary fields inserted at u and v yields the scaling dimension Δ_n. The Ward identity for the two-point function $\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle$ is:

$$\langle T(z)\mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = \left[\frac{h_n}{(z-u)^2} + \frac{h_n}{(z-v)^2} + \frac{1}{z-u}\partial_u + \frac{1}{z-v}\partial_v\right]\langle \mathcal{T}_n\tilde{\mathcal{T}}_n\rangle. \tag{260}$$

Since $\langle T(z)\rangle_{\mathcal{R}_n}$ is precisely the normalized one-point function of T in the presence of the twist fields (i.e., $\langle T(z)\mathcal{T}_n\tilde{\mathcal{T}}_n\rangle/\langle \mathcal{T}_n\tilde{\mathcal{T}}_n\rangle$), we can match the coefficients of the double poles to obtain:

$$h_n = \frac{c}{24}\left(n - \frac{1}{n}\right), \tag{261}$$

and similarly for $\bar{h}_n$, giving $\Delta_n = 2h_n$.

This confirms that the twist fields are primary with the given dimension, and hence the partition function ratio equals their two-point function.

Geometric Picture	**Operator Picture**
Z_n: partition function on $\mathcal{R}_n$	n decoupled copies on $\mathbb{C}$
Branch cut along $[u, v]$	Twist fields at u and v
Cyclic gluing of sheets	Cyclic permutation of copies
$Z_n/(Z_1)^n$: normalized ratio	$\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle$: correlator

Table 10: Equivalence between geometric and operator formulations of the replica trick.

Summary:

- **Geometric picture:** Z_n is the partition function on the multi-sheeted surface $\mathcal{R}_n$
- **Operator picture:** The same path integral can be reinterpreted as n decoupled copies on the plane, with boundary conditions across the cut enforced by twist fields inserted at the branch points
- **Normalization:** Dividing by $(Z_1)^n$ yields the normalized correlation function of twist fields
- **Conformal symmetry:** As primary fields, their two-point function is fixed up to a constant, giving the scaling behavior $Z_n/(Z_1)^n \sim |u-v|^{-2\Delta_n}$

Thus, the relationship between the partition function and the primary field average is not an assumption—it is a direct consequence of the path integral representation of operator insertions as boundary conditions.

4.11 Operator Formalism of the Replica Trick

We describe the operator formalism of the replica trick, defining every object and explaining the logical flow from geometric construction to correlation functions.

4.11.1 The Geometric Setup: $\mathcal{R}_{n,1}$

We consider a single interval $A = [u, v]$ on the real axis of a 1+1 dimensional Euclidean spacetime (complex plane with coordinate $w = x + i\tau$). The replica trick instructs us to consider an n-sheeted Riemann surface $\mathcal{R}_{n,1}$, formed by taking n copies of the plane, cutting each along $[u, v]$, and cyclically gluing them.

The subscript 1 in $\mathcal{R}_{n,1}$ stands for "one interval" (as opposed to multiple intervals $\mathcal{R}_{n,N}$ for N disjoint intervals).

4.11.2 Uniformizing Transformation

The goal is to map this complicated multi-sheeted surface to the ordinary complex plane so we can use standard CFT tools. The transformation is done in two steps:

Step 1: Möbius Map

$$\zeta = \frac{w - u}{w - v}. \tag{262}$$

This is a fractional linear (Möbius) transformation with the following properties:

- It sends the branch point $w = u$ to $\zeta = 0$.
- It sends the other branch point $w = v$ to $\zeta = \infty$.
- The cut interval $[u, v]$ on the real axis is mapped to the negative real axis (or a ray) in the ζ-plane.

Step 2: Power Map (Unwrapping)

$$z = \zeta^{1/n} = \left(\frac{w - u}{w - v}\right)^{1/n}. \tag{263}$$

This takes the n sheets (which are wrapped around the branch points) and unwraps them into a single complex z-plane. A full turn around u (or v) on $\mathcal{R}_{n,1}$ corresponds to an angle increase of $2\pi n$ in the w-plane, which becomes $2\pi n/n = 2\pi$ in the z-plane. Hence the map is single-valued and bijective from $\mathcal{R}_{n,1}$ to $\mathbb{C}$.

w**-plane**	ζ**-plane**	z**-plane**
$\mathcal{R}_{n,1}$ (n sheets)	Intermediate (branch points at 0, ∞)	$\mathbb{C}$ (single sheet)
Branch points: u, v	$\zeta(u) = 0$, $\zeta(v) = \infty$	Unwrapped by $z = \zeta^{1/n}$
Cut along $[u, v]$	Cut along negative real axis	No cut (smooth plane)

Table 11: Two-step uniformizing transformation from $\mathcal{R}_{n,1}$ to $\mathbb{C}$.

4.11.3 Operator Identification

The fundamental relation between correlation functions on the curved space $\mathcal{R}_{n,1}$ and normalized correlation functions of twist fields on the plane is:

$$\langle O(x,\tau;\text{sheet } i)\cdots\rangle_{\mathcal{L},\mathcal{R}_{n,1}} = \frac{\langle \mathcal{T}_n(u_1,0)\tilde{\mathcal{T}}_n(v_1,0)O_i(x,\tau)\cdots\rangle_{\mathcal{L}^{(n)},\mathbb{C}}}{\langle \mathcal{T}_n(u_1,0)\tilde{\mathcal{T}}_n(v_1,0)\rangle_{\mathcal{L}^{(n)},\mathbb{C}}}. \tag{17}$$

Definition of symbols:

- $\mathcal{L}$: The original Lagrangian of the CFT (e.g., a free boson or Ising model).
- $\mathcal{R}_{n,1}$: The n-sheeted Riemann surface for one interval.
- $\langle\cdots\rangle_{\mathcal{L},\mathcal{R}_{n,1}}$: The path integral expectation value on that curved space with the original action.
- $O(x,\tau;\text{sheet } i)$: A local operator (e.g., a primary field) inserted at coordinates (x,τ) on the i-th sheet of the multi-sheeted surface.
- $\mathcal{L}^{(n)}$: The theory of n independent copies of the original CFT. Its fields are $\phi_1,\phi_2,\ldots,\phi_n$, each with action $S[\phi_j]$. These copies are decoupled in the bulk; the only coupling comes from the twist operators.
- $\mathcal{T}_n(u,0)$: The twist field inserted at the branch point u (with $\tau=0$ because the cut is on the real axis). Its job is to implement the cyclic gluing: when any field is transported around u, it moves from copy j to copy $j+1$.
- $\tilde{\mathcal{T}}_n(v,0)$: The anti-twist field at v. It implements the inverse cyclic permutation, closing the loop. (Sometimes called $\bar{\mathcal{T}}_n$ or $\mathcal{T}_{-n}$.)
- $O_i(x,\tau)$: The same operator O, but now inserted in the i-th copy of the orbifold theory on the ordinary complex plane $\mathbb{C}$.
- **Denominator**: $\langle \mathcal{T}_n\tilde{\mathcal{T}}_n\rangle_{\mathcal{L}^{(n)},\mathbb{C}}$ is the vacuum expectation value of the two twist fields on the plane. It provides the correct normalization.

Why is this equality true?

The path integral on the curved space $\mathcal{R}_{n,1}$ can be "flattened" by the uniformizing map $w\mapsto z$. In the operator language, this flattening corresponds to replacing the nontrivial geometry with a flat plane plus boundary condition changing operators at the branch points. The twist field $\mathcal{T}_n(u)$ creates a branch cut such that fields acquire a monodromy $j\to j+1$ around u. The anti-twist $\tilde{\mathcal{T}}_n(v)$ closes the cut. Therefore, doing the path integral on the plane with these two insertions is exactly equivalent to doing it on $\mathcal{R}_{n,1}$.

The division by $\langle \mathcal{T}_n\tilde{\mathcal{T}}_n\rangle$ is necessary because the left-hand side is a standard expectation value (normalized by the partition function on $\mathcal{R}_{n,1}$), while the numerator on the right is an unnormalized path integral with insertions. The denominator provides the missing normalization factor.

4.11.4 Physical Interpretation of the Gluing

The twist fields enforce specific sewing conditions:

- **On the interval $A = [u, v]$ (the cut):** The twist fields force the fields at $\tau = 0^+$ (just above the cut) on copy j to equal the fields at $\tau = 0^-$ (just below the cut) on copy $j + 1$. This is the *cyclic gluing*.
- **On the complement B (outside the interval):** There is no cut, so the twist fields together impose periodic boundary conditions: $\phi_j(\tau = 0^+) = \phi_j(\tau = 0^-)$. The copies are sewn to themselves.

Thus, for $x \in [u_j, v_j]$, consecutive copies are connected through $\tau = 0$ due to the presence of $\mathcal{T}_n(v_j, 0)$, whereas for x in B, copies are connected to themselves through $\tau = 0$ because the conditions arising from the definition of $\mathcal{T}_n(u_j, 0)$ and $\tilde{\mathcal{T}}_n(v_j, 0)$ cancel each other.

4.11.5 Diagonalizing the Twist: Fourier Transform

For free field theories, the action is quadratic, and the cyclic coupling introduced by the twist fields can be diagonalized by a discrete Fourier transform.

Define new fields for each copy $j = 1, \ldots, n$:

$$\tilde{\varphi}_k = \sum_{j=1}^{n} e^{2\pi i k j/n} \varphi_j, \qquad k = 0, 1, \ldots, n-1. \tag{18}$$

Here:

- φ_j is the original field on the j-th copy.
- $\tilde{\varphi}_k$ are complex linear combinations (Fourier modes).

The twist operator $\mathcal{T}_n$ acts by cyclically permuting the copies: $\mathcal{T}_n \varphi_j = \varphi_{j+1}$ (with $\varphi_{n+1} \equiv \varphi_1$). Let's see how $\tilde{\varphi}_k$ transforms:

$$\mathcal{T}_n \tilde{\varphi}_k = \sum_{j=1}^{n} e^{2\pi i k j/n} \varphi_{j+1} \tag{264}$$

$$= e^{-2\pi i k/n} \sum_{j=1}^{n} e^{2\pi i k (j+1)/n} \varphi_{j+1} \tag{265}$$

$$= e^{-2\pi i k/n} \tilde{\varphi}_k. \tag{266}$$

The anti-twist $\tilde{\mathcal{T}}_n$ gives the opposite phase $e^{2\pi i k/n}$.

Important: The fields $\tilde{\varphi}_k$ are *eigenfields* of the twist operation. Moreover, because the action is quadratic and diagonal in the Fourier basis, the different k modes decouple from each other. The total action becomes a sum of independent actions for each $\tilde{\varphi}_k$.

For a real field φ_j, we have the reality condition $\tilde{\varphi}_k^* = \tilde{\varphi}_{n-k}$, so the independent modes are only for $k = 0, 1, \ldots, \lfloor n/2 \rfloor$, with appropriate reality constraints.

4.11.6 Factorizing the Twist Fields

Since the k-modes are decoupled, the twist field itself factorizes into a product of operators that each act on a single mode:

$$\mathcal{T}_n = \prod_{k=0}^{n-1} \mathcal{T}_{n,k}, \qquad \tilde{\mathcal{T}}_n = \prod_{k=0}^{n-1} \tilde{\mathcal{T}}_{n,k}. \tag{20}$$

Here $\mathcal{T}_{n,k}$ is the twist operator for the k-th mode. It multiplies $\tilde{\varphi}_k$ by $e^{\pm 2\pi i k/n}$ and leaves other modes untouched.

4.11.7 Factorizing the Partition Function

Because both the action and the twist fields factorize, the partition function on the replica surface factorizes as well:

$$Z_{\mathcal{R}} = \prod_{k=0}^{n-1} \langle \mathcal{T}_{k,n}(u,0) \tilde{\mathcal{T}}_{k,n}(v,0) \cdots \rangle_{\mathcal{L}^{(n)},\mathbb{C}}. \tag{21}$$

Each factor is a correlation function of the twist fields for a single decoupled mode k. For a free boson or fermion, each factor can be computed exactly using Green's functions on the complex plane with a conical singularity. The product over k then gives the full partition function ratio $Z_n/(Z_1)^n$, and ultimately the entanglement entropy.

Original Basis	**Fourier Basis**
Fields: $\varphi_1, \ldots, \varphi_n$	Modes: $\tilde{\varphi}_0, \ldots, \tilde{\varphi}_{n-1}$
Twist: cyclic permutation $j \to j+1$	Twist: phase multiplication $e^{-2\pi i k/n}$
Coupled through boundary conditions	Decoupled (diagonal action)
Single twist field $\mathcal{T}_n$	Product $\prod_k \mathcal{T}_{n,k}$

Table 12: Diagonalization of the replica trick via discrete Fourier transform.

Summary of Logical Steps:

1. **Uniformizing map:** $w \mapsto \zeta = (w-u)/(w-v) \mapsto z = \zeta^{1/n}$ flattens $\mathcal{R}_{n,1}$ to the plane.
2. **Operator dictionary:** Correlation functions on the curved space equal normalized correlation functions of twist fields on the plane (Eq. 17). This is the fundamental bridge between geometry and algebra.
3. **Free theory trick:** Diagonalize the cyclic twist by taking discrete Fourier transforms of the n replica fields (Eqs. 18–19).
4. **Decoupling:** The action and twist fields factor into independent k-modes (Eq. 20).
5. **Factorization of Z_n:** The partition function becomes a product of n two-point functions (Eq. 21), enabling explicit calculations.

This machinery is the backbone of all exact entanglement entropy calculations in free 2D CFTs and underpins the universal results for interacting CFTs via twist field scaling dimensions.

4.12 Two-Point Function of Twist Fields

For primary fields with scaling dimension Δ_n on the plane:

$$\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = \frac{c_n}{|u-v|^{2\Delta_n}} \tag{267}$$

where c_n is a non-universal constant.

Substitute Δ_n:

$$\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = c_n|u-v|^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} \tag{268}$$

Relate to partition function:

$$\frac{Z_n}{(Z_1)^n} = \langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = c_n\left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} \tag{269}$$

where $L = |u-v|$ and a is the UV cutoff (lattice spacing).

5 The $n \to 1$ Limit

5.1 Rényi Entropy

From Eq. (52):

$$\mathrm{Tr}(\rho_A^n) = c_n \left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} \tag{270}$$

Rényi entropy:

$$S_A^{(n)} = \frac{1}{1-n} \ln \mathrm{Tr}(\rho_A^n) \tag{271}$$

$$= \frac{1}{1-n}\left[\ln c_n - \frac{c}{6}\left(n - \frac{1}{n}\right)\ln\left(\frac{L}{a}\right)\right] \tag{272}$$

$$= \frac{c}{6}\left(1+\frac{1}{n}\right)\ln\left(\frac{L}{a}\right) + \frac{\ln c_n}{1-n} \tag{273}$$

Define $c_n' = \frac{\ln c_n}{1-n}$ (non-universal constant).

5.2 Von Neumann Entropy

Take the limit $n \to 1$:

$$S_A = \lim_{n\to 1} S_A^{(n)} = \frac{c}{3}\ln\left(\frac{L}{a}\right) + c_1' \tag{274}$$

Alternative derivation via L'Hôpital's rule:
From the replica trick formula:

$$S_A = -\left.\frac{\partial}{\partial n}\mathrm{Tr}(\rho_A^n)\right|_{n=1} \tag{275}$$

Compute the derivative:

$$\frac{\partial}{\partial n}\mathrm{Tr}(\rho_A^n) = \frac{\partial}{\partial n}\left[c_n\left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)}\right] \tag{276}$$

$$= \left(\frac{\partial c_n}{\partial n}\right)\left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} + c_n\left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} \cdot \left(-\frac{c}{6}\right)\left(1+\frac{1}{n^2}\right)\ln\left(\frac{L}{a}\right) \tag{277}$$

Evaluate at $n = 1$:

$$\left.\frac{\partial}{\partial n}\mathrm{Tr}(\rho_A^n)\right|_{n=1} = \left.\frac{\partial c_n}{\partial n}\right|_{n=1} - \frac{c}{3}\ln\left(\frac{L}{a}\right) \tag{278}$$

Therefore:

$$S_A = -\left.\frac{\partial c_n}{\partial n}\right|_{n=1} + \frac{c}{3}\ln\left(\frac{L}{a}\right) = \frac{c}{3}\ln\left(\frac{L}{a}\right) + \mathrm{const} \tag{279}$$

5.3 Main Result

Entanglement Entropy in 1+1D CFT
For a subsystem of length L in the vacuum state of a 1+1D CFT:

$$S_A = \frac{c}{3} \ln \left(\frac{L}{a} \right) + \text{const} \tag{280}$$

where:

- c: central charge of the CFT
- L: length of subsystem A
- a: UV cutoff (lattice spacing)

6 Numerical Method

6.1 Correlation Matrix Approach

For free fermion systems (e.g., transverse Ising model), the entanglement entropy can be computed from the correlation matrix.

Correlation matrix:

$$C_{ij} = \langle c_i^\dagger c_j \rangle \quad \text{for } i, j \in A \tag{281}$$

Entanglement entropy:

$$S_A = -\operatorname{Tr}\left[C \ln C + (I - C)\ln(I - C)\right] \tag{282}$$

where the trace is over the subsystem A.

Scaling with subsystem size:

$$S_A(L) = \frac{c}{3} \ln\left(\frac{L}{a}\right) + \text{const} \tag{283}$$

Extracting central charge: Fit numerical data to:

$$S_A(L) = \frac{c}{3} \log_2\left(\frac{L}{a}\right) + \text{const} \tag{284}$$

The slope gives $c/3$, so $c = 3 \times \text{slope}$.

For transverse Ising model at critical point ($g = J$):

- Expected: $c = 1/2$
- Expected slope: $c/3 = 1/6 \approx 0.1667$

Summary

6.2 Key Equations

Equation	Formula	Meaning
Path integral	$\langle\phi_f\|e^{-\beta H}\|\phi_i\rangle = \int \mathcal{D}\phi\, e^{-S_E}$	Transition amplitude
Density matrix	$\rho(\phi_f\|\phi_i) = \frac{1}{Z}\int \mathcal{D}\phi\, e^{-S_E}$	Thermal state
Reduced ρ_A	Cylinder with cuts along A	Partial trace
Replica trick	$\mathrm{Tr}(\rho_A^n) = Z_n(A)/Z^n$	n-sheeted surface
Twist dimension	$\Delta_n = \frac{c}{12}(n - 1/n)$	From stress tensor
Two-point function	$\langle\mathcal{T}_n\tilde{\mathcal{T}}_n\rangle = L^{-2\Delta_n}$	CFT on plane
Rényi entropy	$S_A^{(n)} = \frac{c}{6}(1 + 1/n)\ln(L/a)$	Before $n \to 1$ limit
Von Neumann	$S_A = \frac{c}{3}\ln(L/a)$	Final result

6.3 Derivation Steps

Step	Operation	Result
1	Trotter decomposition	$e^{-\beta H} = (e^{-\epsilon H})^N$
2	Complete sets of states	$\int dq_k\|q_k\rangle\langle q_k\|$
3	Momentum eigenstates	$\langle q\|p\rangle = e^{ipq}$
4	Gaussian integrals	Kinetic term $\frac{1}{2}m\dot{q}^2$
5	Continuum limit	Path integral $\int \mathcal{D}\phi\, e^{-S_E}$
6	Delta functions	Boundary conditions
7	Partial trace	Cylinder with cuts
8	Replica trick	n-sheeted surface
9	Twist fields	CFT correlators
10	$n \to 1$ limit	$S_A = \frac{c}{3}\ln(L/a)$

6.4 Generalizations

Situation	Formula
Infinite system, $T = 0$	$S = \frac{c}{3}\ln(L/a)$
Finite temperature	$S = \frac{c}{3}\ln\left[\frac{\beta}{\pi a}\sinh\left(\frac{\pi L}{\beta}\right)\right]$
Finite size (periodic)	$S = \frac{c}{3}\ln\left[\frac{L}{\pi a}\sin\left(\frac{\pi\ell}{L}\right)\right]$
After global quench	$S(t) \sim \frac{\pi c}{3\beta}t$ (linear growth)
KZM quench	$S_\infty = \frac{c}{6}\ln(\tau_Q \ln \tau_Q)$

Acknowledgments

This document follows the Calabrese-Cardy derivation of entanglement entropy in CFT. For further reading:

- Calabrese & Cardy (2004). "Entanglement entropy and quantum field theory" J. Stat. Mech. P06002
- Calabrese & Cardy (2009). "Entanglement entropy and conformal field theory" J. Phys. A 42 504005
- Holzhey, Larsen, Wilczek (1994). "Geometric and renormalized entropy in conformal field theory" Nucl. Phys. B424 443

Appendix A: Position-Momentum Overlap

Goal: Derive $\langle p|q\rangle = e^{-ipq}$ from first principles.

Conventions

We work in units where $\hbar = 1$. The position and momentum operators satisfy:

$$[\hat{q}, \hat{p}] = i \tag{285}$$

Position eigenstates:

$$\hat{q}|q\rangle = q|q\rangle, \quad \langle q|q'\rangle = \delta(q - q') \tag{286}$$

Momentum eigenstates:

$$\hat{p}|p\rangle = p|p\rangle, \quad \langle p|p'\rangle = 2\pi\delta(p - p') \tag{287}$$

The 2π factor is a convention choice (alternative: $\langle p|p'\rangle = \delta(p - p')$ with different Fourier conventions).

Derivation

Step 1: Translation operator

The momentum operator generates translations in position:

$$e^{-ia\hat{p}}|q\rangle = |q + a\rangle \tag{288}$$

Proof: Apply $\hat{q}$ to both sides:

$$\begin{aligned}
\hat{q}e^{-ia\hat{p}}|q\rangle &= e^{-ia\hat{p}}(e^{ia\hat{p}}\hat{q}e^{-ia\hat{p}})|q\rangle && (289)\\
&= e^{-ia\hat{p}}(\hat{q} + a)|q\rangle \quad \text{(using } [\hat{q}, \hat{p}] = i) && (290)\\
&= e^{-ia\hat{p}}(q + a)|q\rangle && (291)\\
&= (q + a)e^{-ia\hat{p}}|q\rangle && (292)
\end{aligned}$$

Thus $e^{-ia\hat{p}}|q\rangle$ is an eigenstate of $\hat{q}$ with eigenvalue $q + a$, so it equals $|q + a\rangle$ (up to phase, which we set to 1).

Step 2: Matrix element of translation

Consider $\langle p|q\rangle$. Apply the translation operator:

$$\begin{aligned}
\langle p|q + a\rangle &= \langle p|e^{-ia\hat{p}}|q\rangle && (293)\\
&= e^{-iap}\langle p|q\rangle \quad \text{(since } \hat{p}|p\rangle = p|p\rangle) && (294)
\end{aligned}$$

Step 3: Differential equation

Expand for infinitesimal a:

$$\langle p|q + a\rangle = \langle p|q\rangle + a\frac{\partial}{\partial q}\langle p|q\rangle + O(a^2) \tag{295}$$

$$e^{-iap}\langle p|q\rangle = (1 - iap + O(a^2))\langle p|q\rangle \tag{296}$$

Matching $O(a)$ terms:

$$\frac{\partial}{\partial q}\langle p|q\rangle = -ip\langle p|q\rangle \tag{297}$$

Step 4: Solve the differential equation

The solution is:

$$\langle p|q\rangle = C(p)e^{-ipq} \tag{298}$$

where $C(p)$ is a q-independent normalization constant.

Step 5: Determine normalization

Use the completeness relation for position states:

$$\langle p|p'\rangle = \int dq\, \langle p|q\rangle\langle q|p'\rangle \tag{299}$$

Substitute $\langle p|q\rangle = C(p)e^{-ipq}$ and $\langle q|p'\rangle = \langle p'|q\rangle^* = C^*(p')e^{ip'q}$:

$$\langle p|p'\rangle = \int dq\, C(p)e^{-ipq}C^*(p')e^{ip'q} \tag{300}$$

$$= C(p)C^*(p')\int dq\, e^{i(p'-p)q} \tag{301}$$

$$= C(p)C^*(p')\cdot 2\pi\delta(p'-p) \tag{302}$$

We require $\langle p|p'\rangle = 2\pi\delta(p-p')$, so:

$$C(p)C^*(p') = 1 \quad \Rightarrow \quad |C(p)| = 1 \tag{303}$$

We choose the phase convention $C(p) = 1$, giving:

$$\boxed{\langle p|q\rangle = e^{-ipq}} \tag{304}$$

Alternative Convention

Some texts use symmetric normalization:

$$\langle p|q\rangle = \frac{1}{\sqrt{2\pi}}e^{-ipq} \tag{305}$$

with $\langle p|p'\rangle = \delta(p-p')$ and completeness $\int dp\, |p\rangle\langle p| = I$.

Our convention ($\langle p|q\rangle = e^{-ipq}$) is common in path integral literature and simplifies formulas by absorbing 2π factors into the momentum measure $\int \frac{dp}{2\pi}$.